Planar Graphs

Alexander Golovnev

Outline

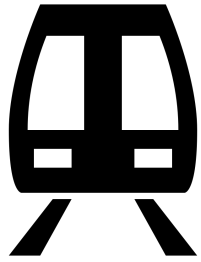
Subway Lines

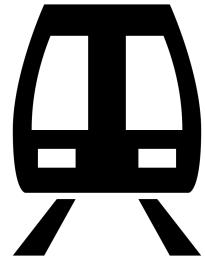
Planar Graphs

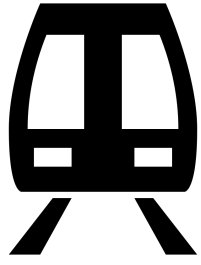
Euler's Formula

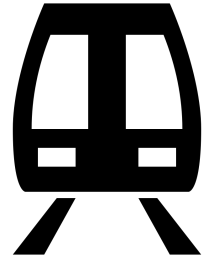
Applications of Euler's Formula

Subway Lines

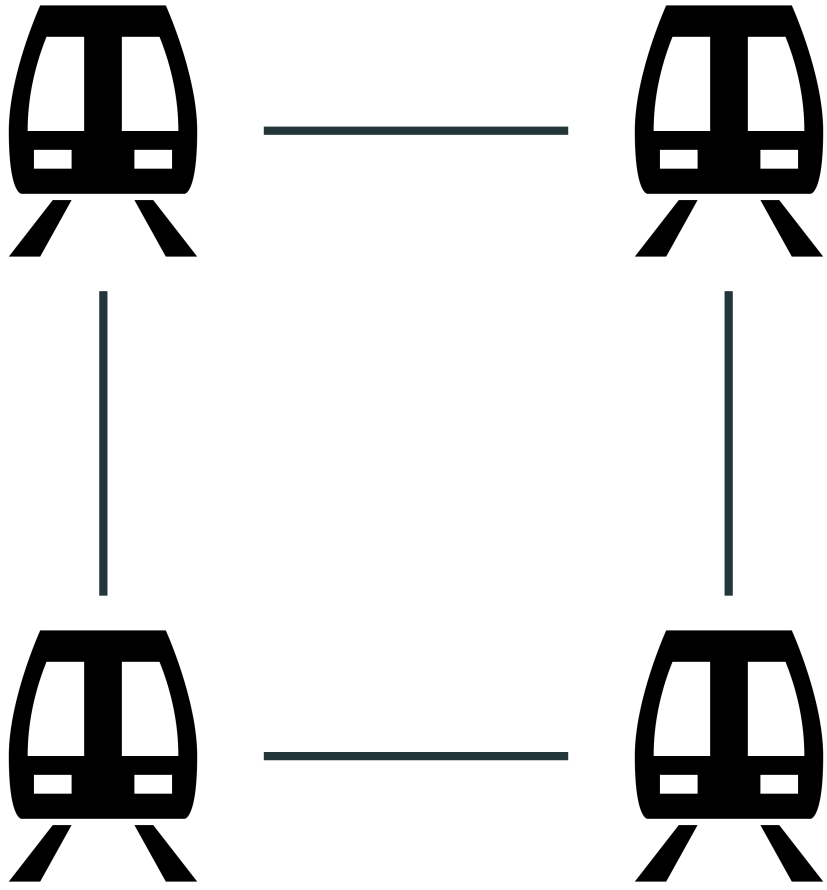




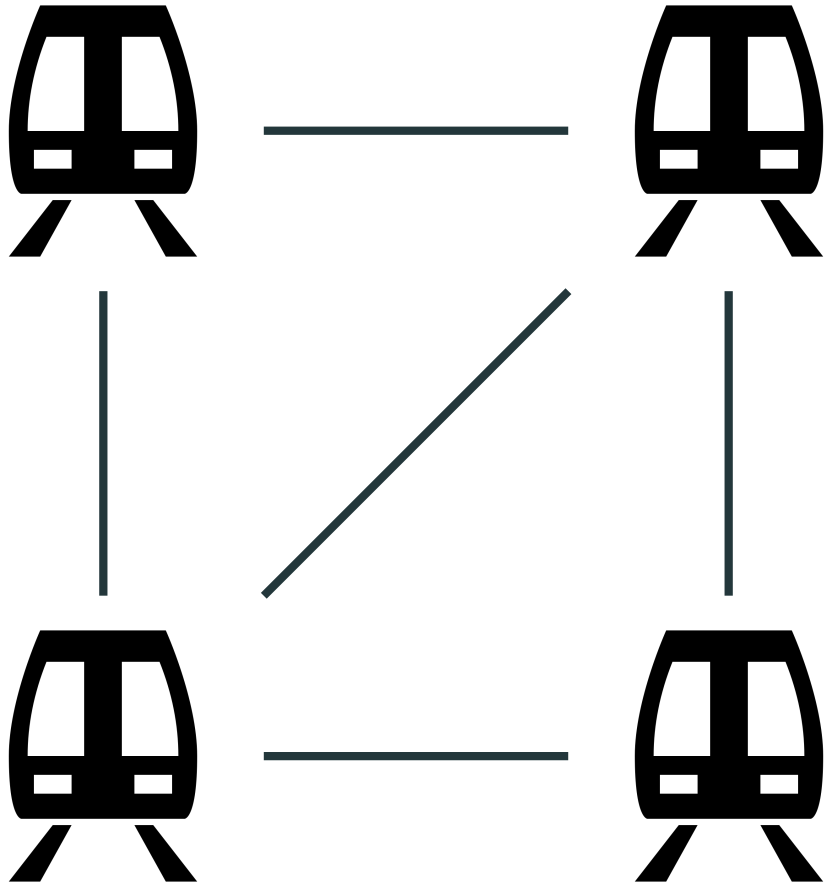




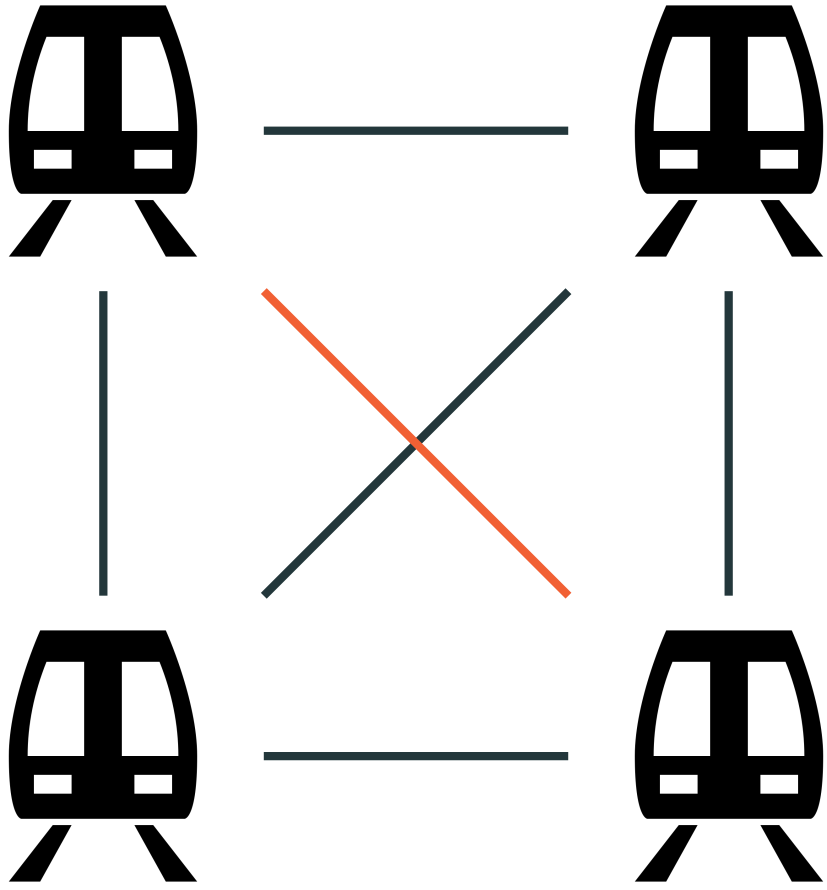
Subway Lines



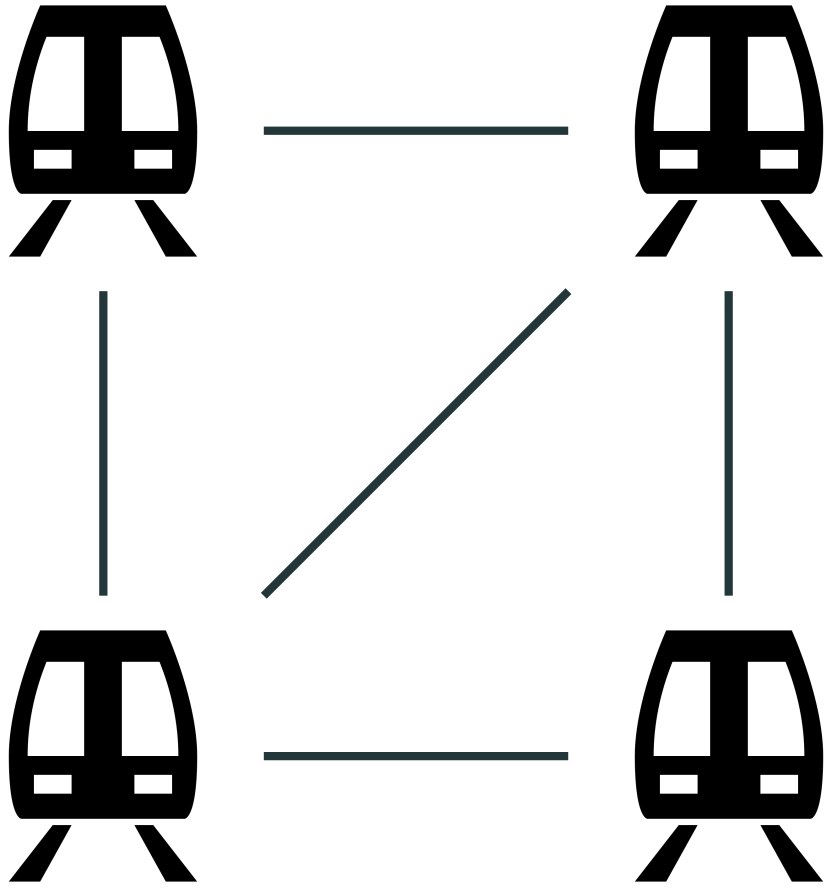
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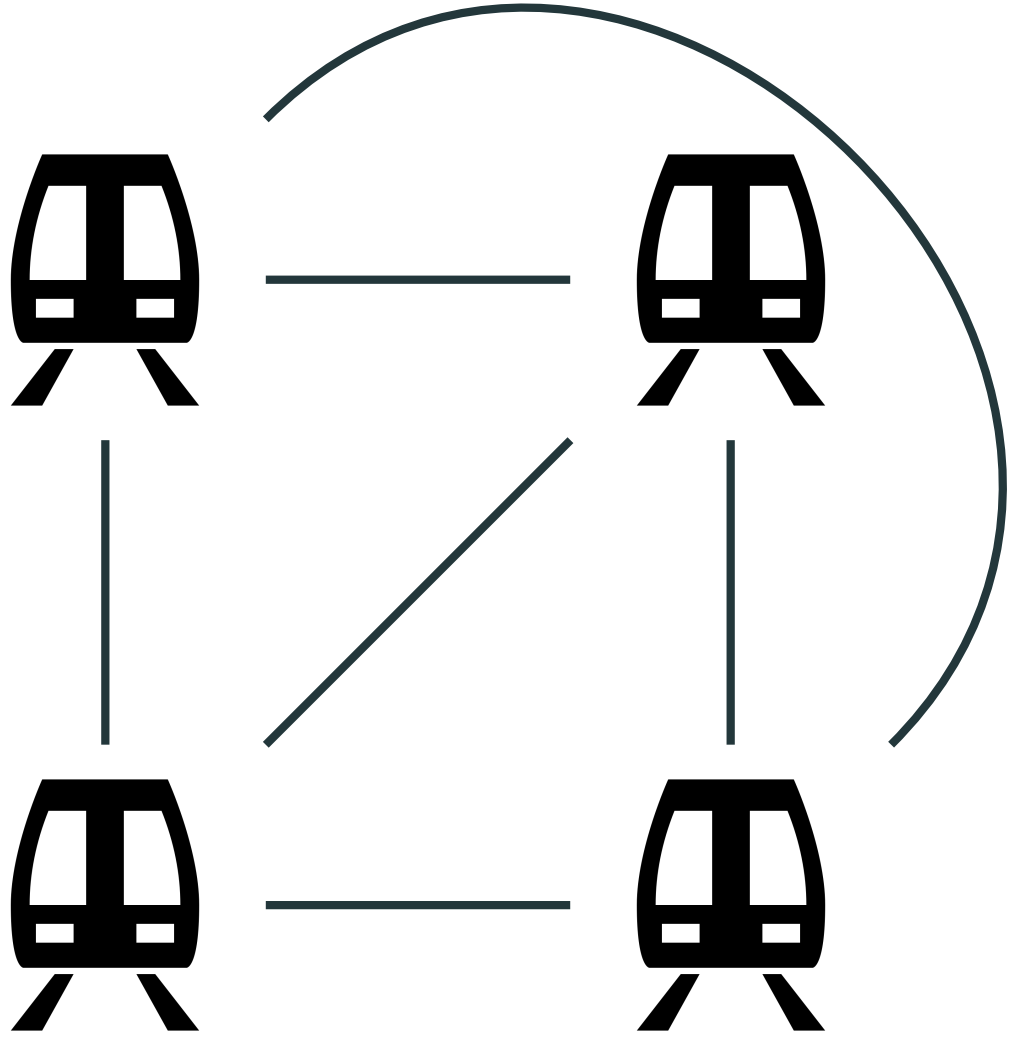
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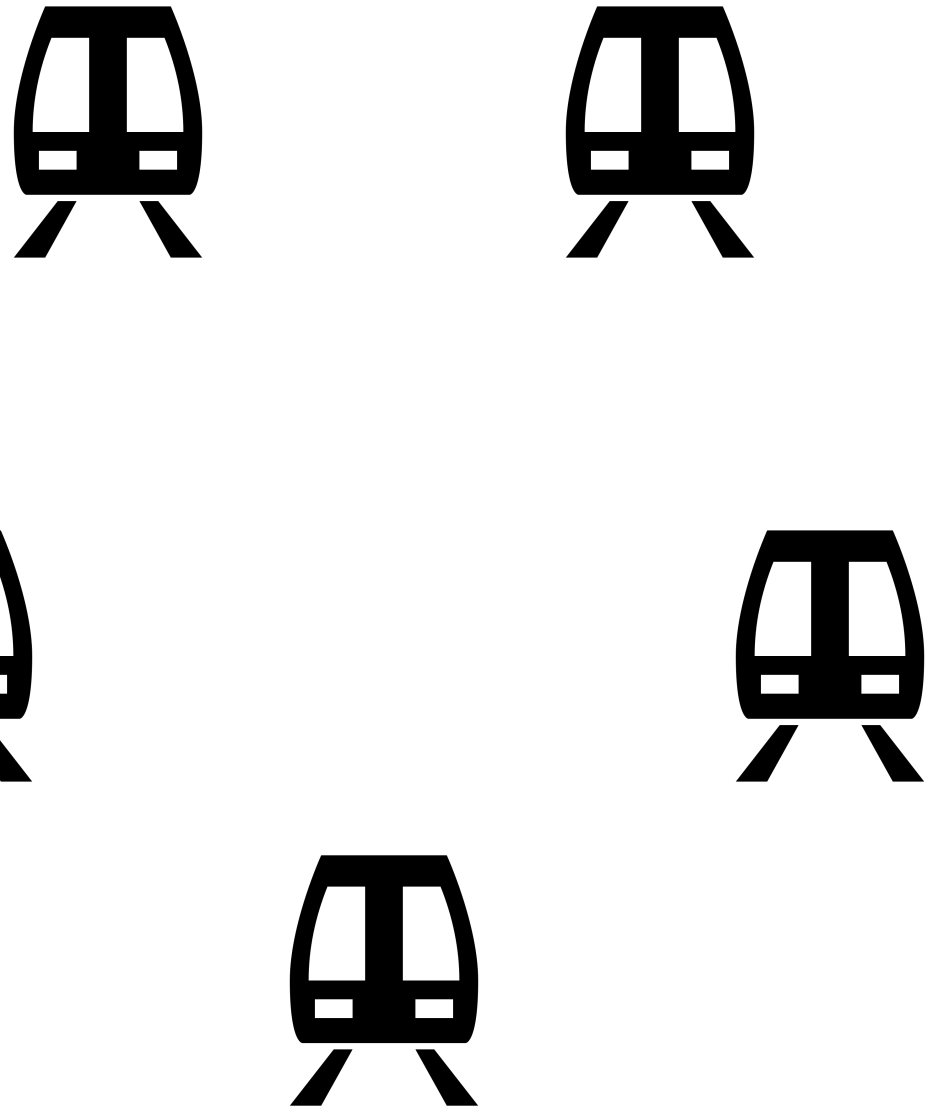
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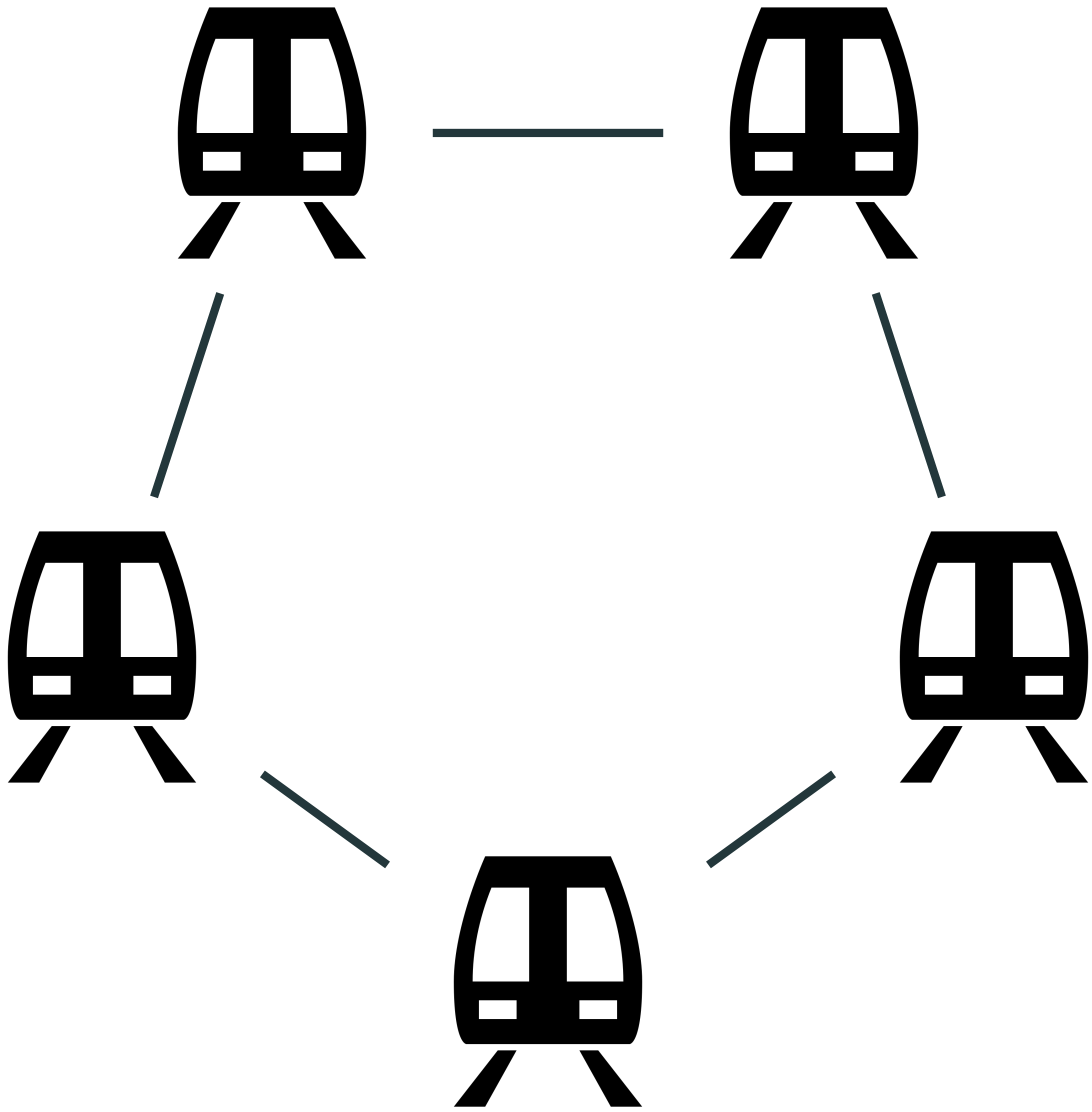
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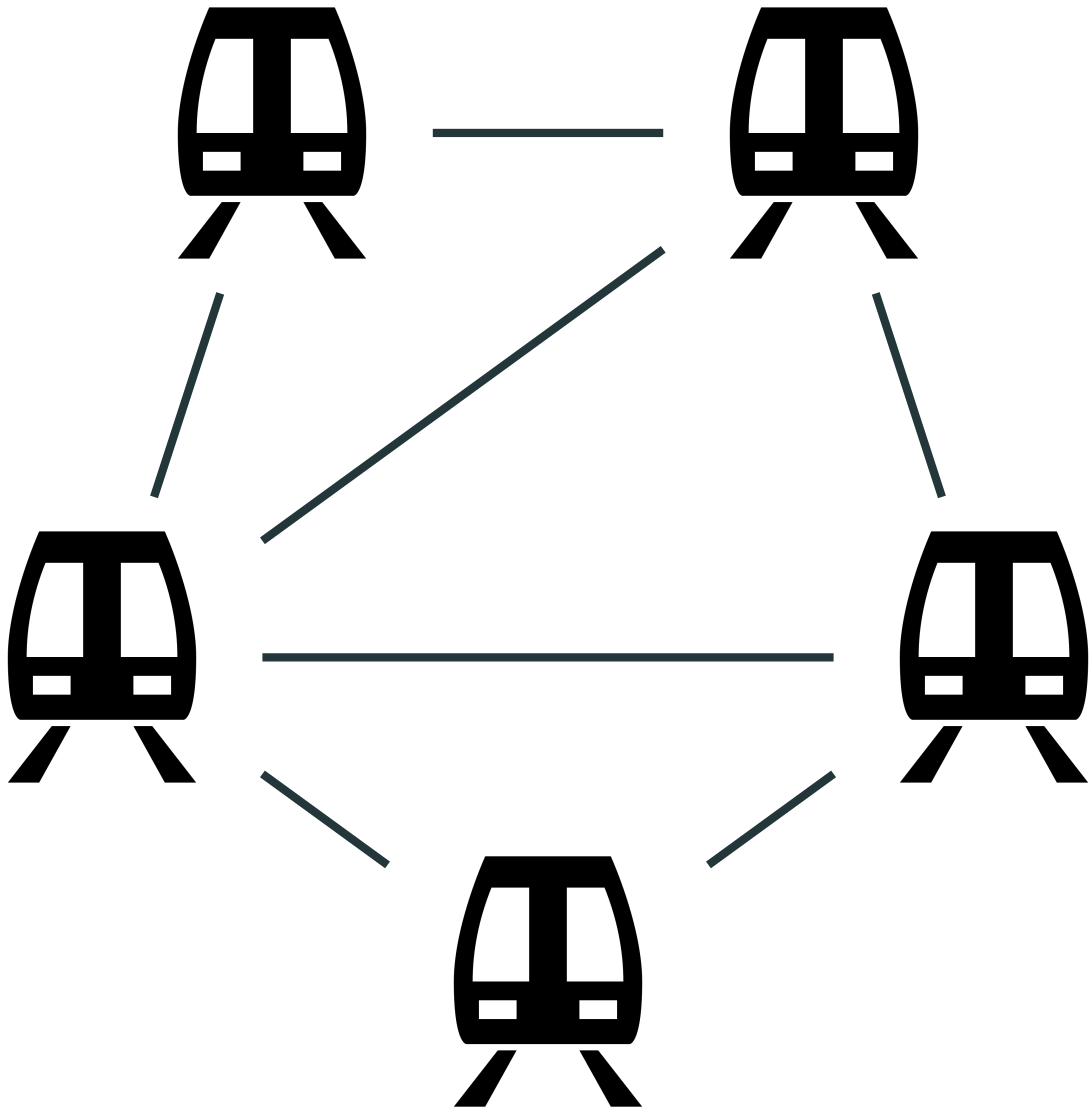
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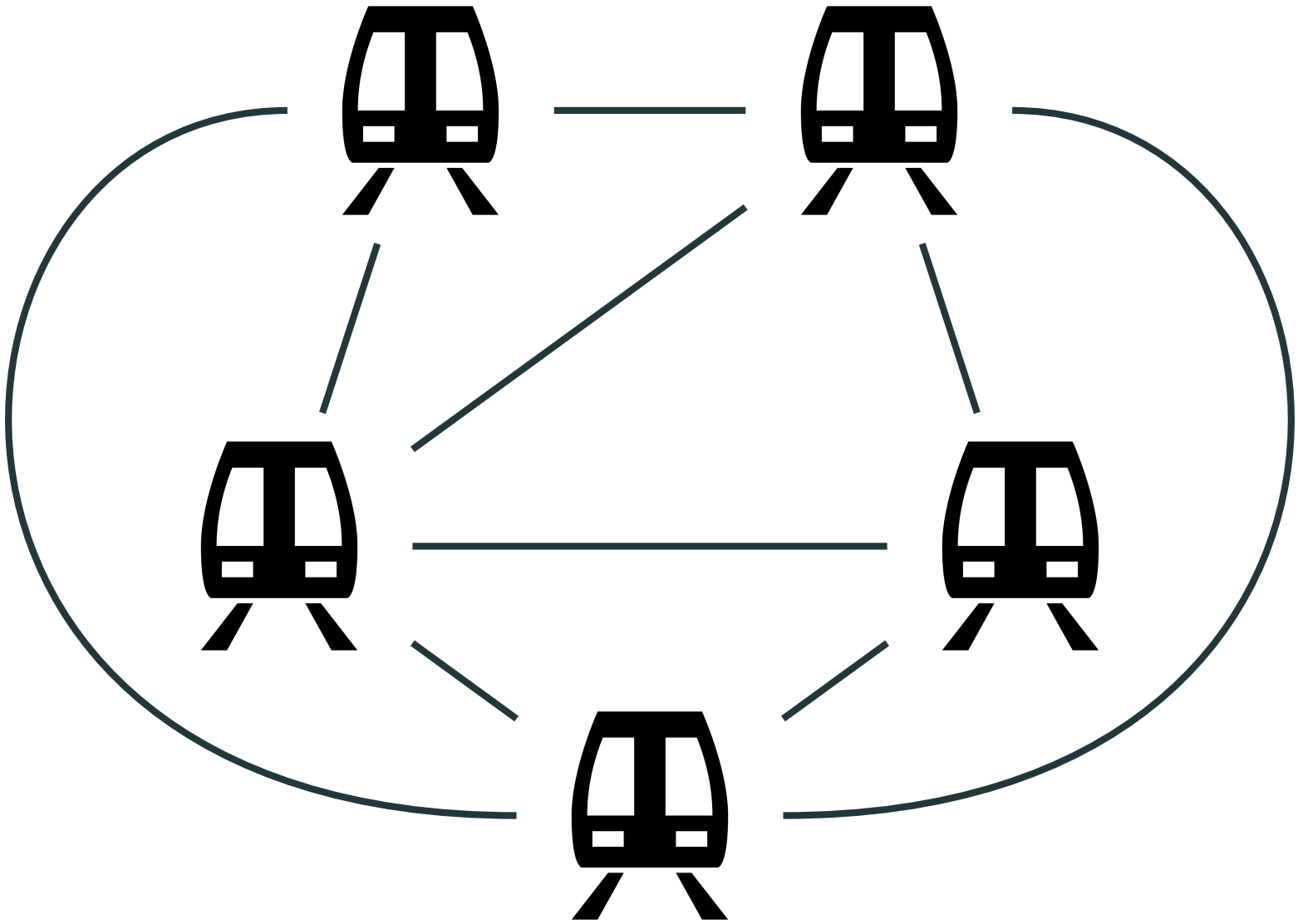
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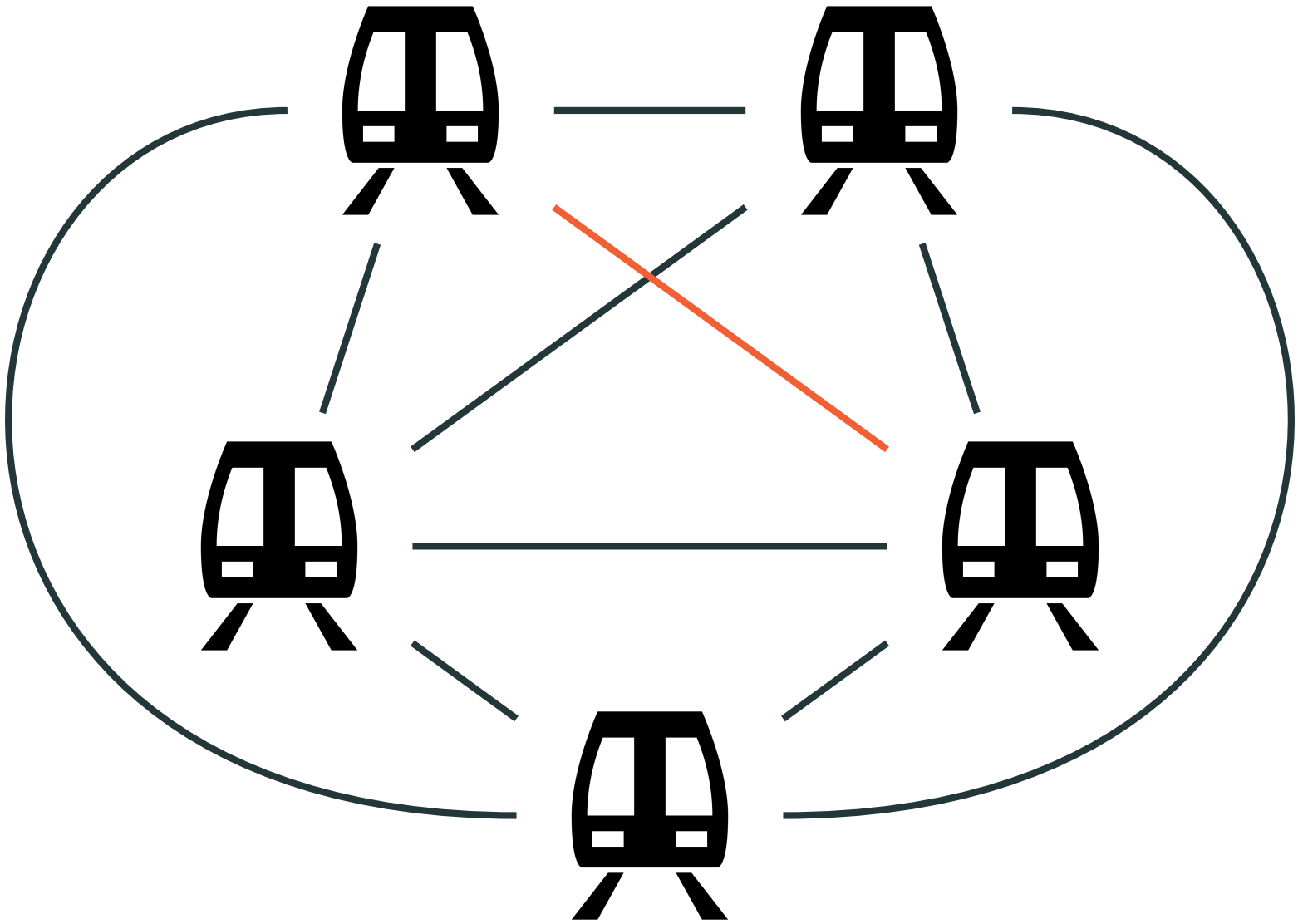
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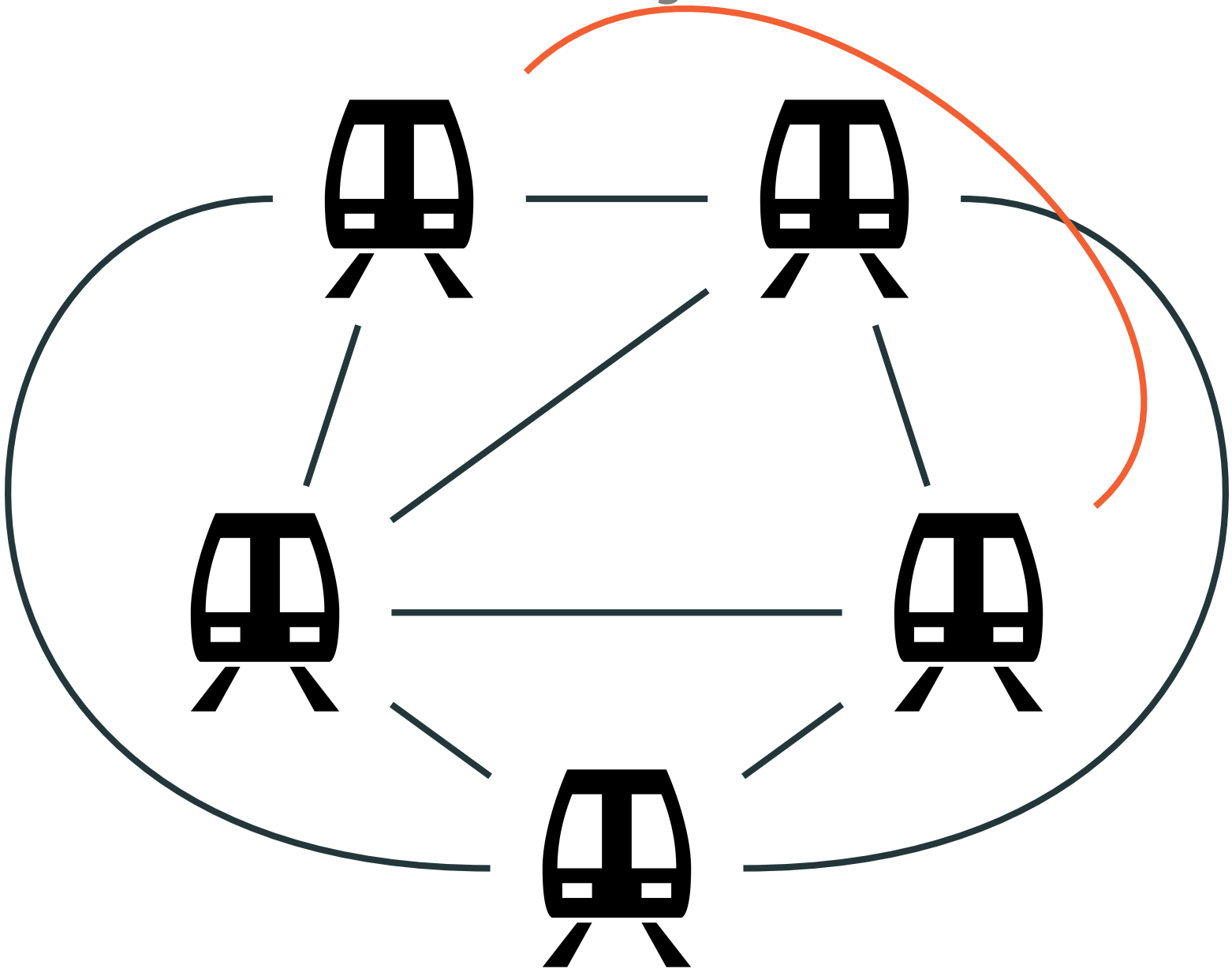
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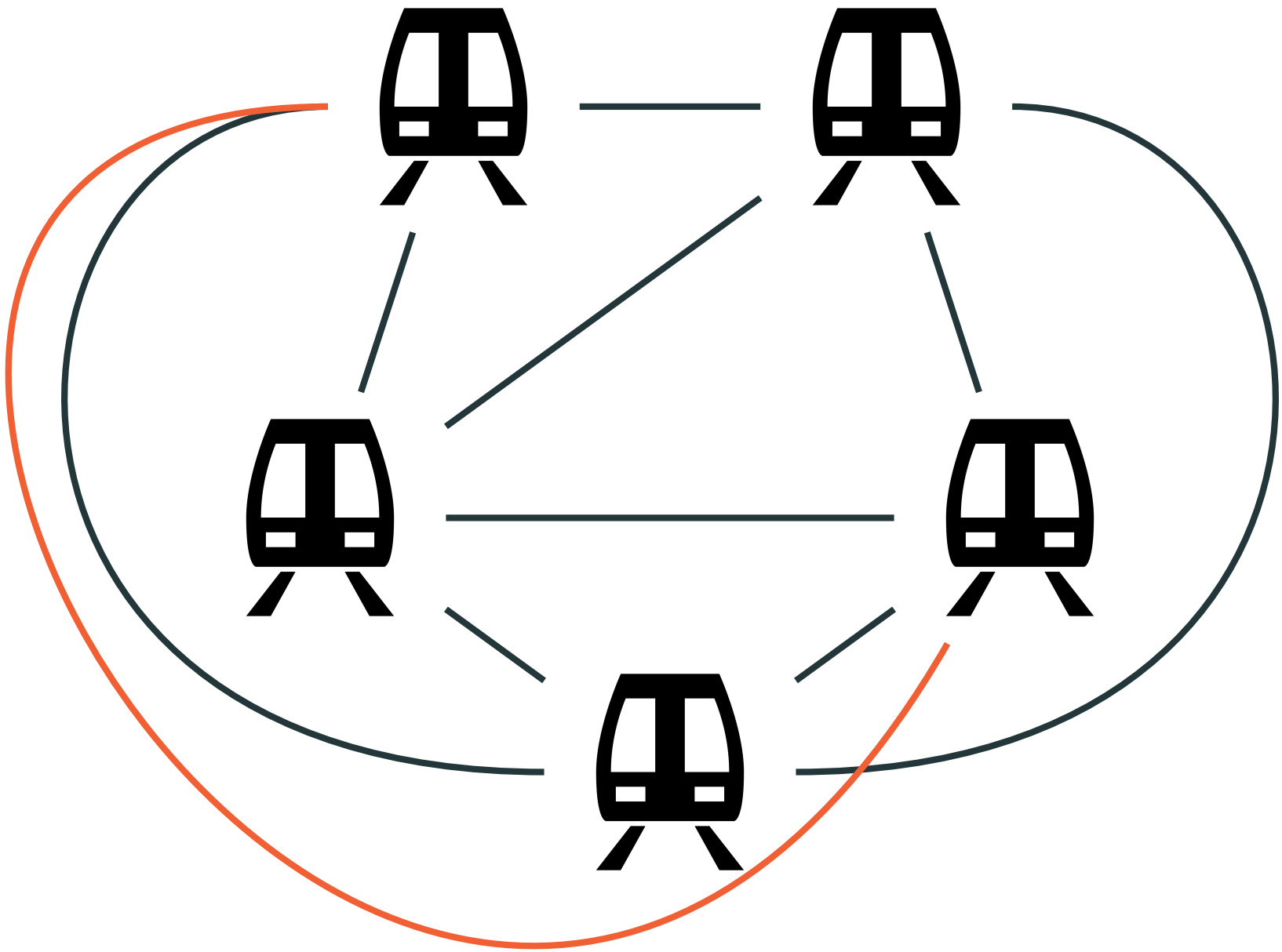
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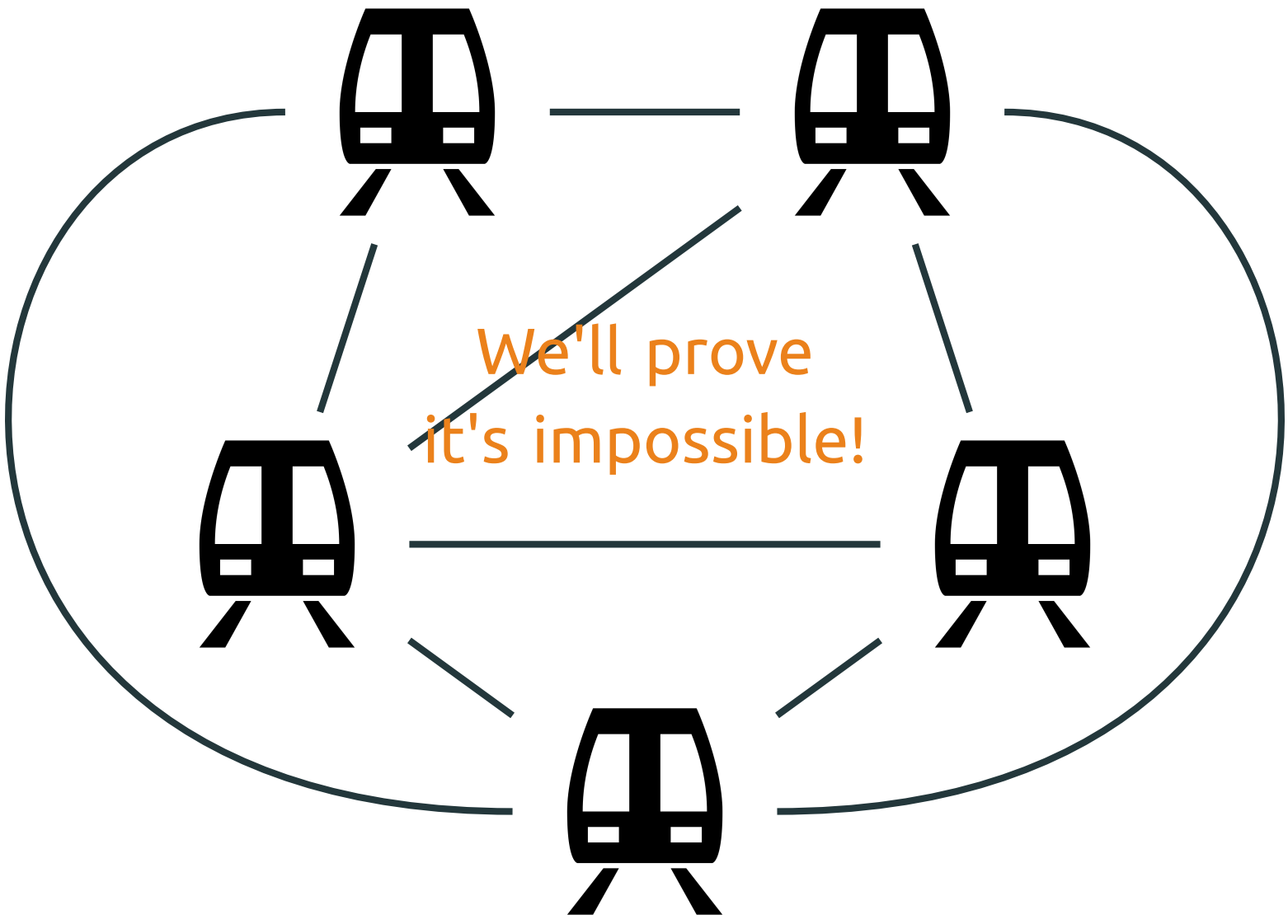
Subway Lines



Subway Lines



Subway Lines



Outline

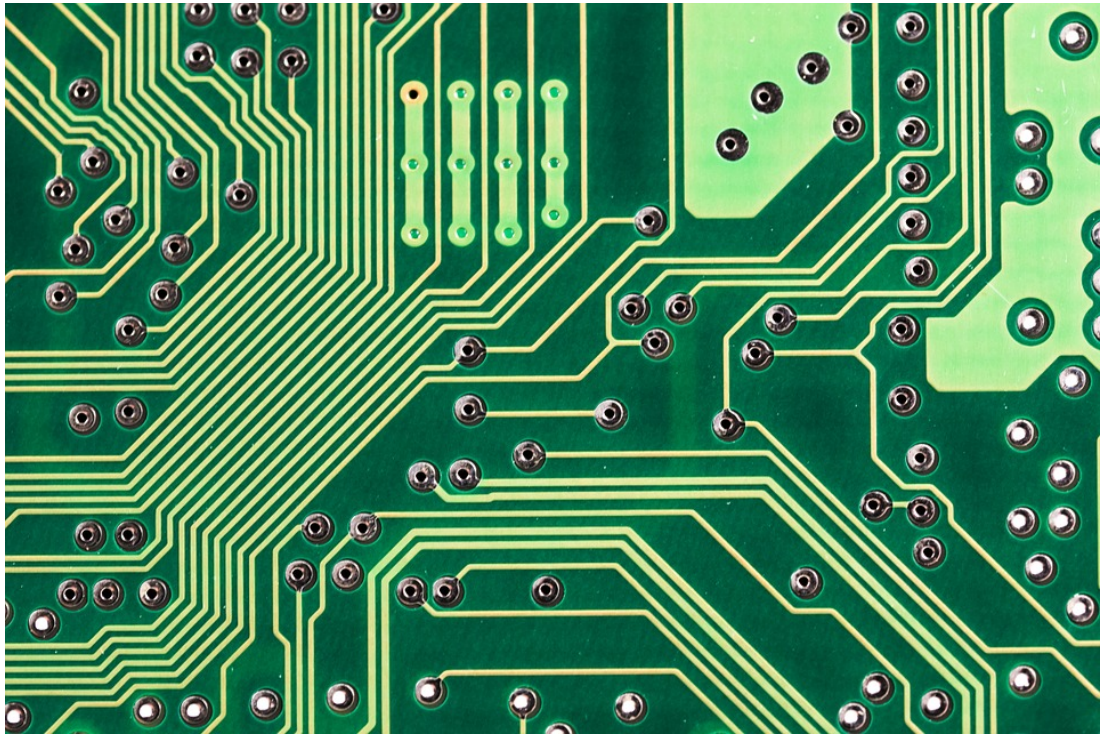
Subway Lines

Planar Graphs

Euler's Formula

Applications of Euler's Formula

Design of Electronic Circuits



Three Utilities Problem





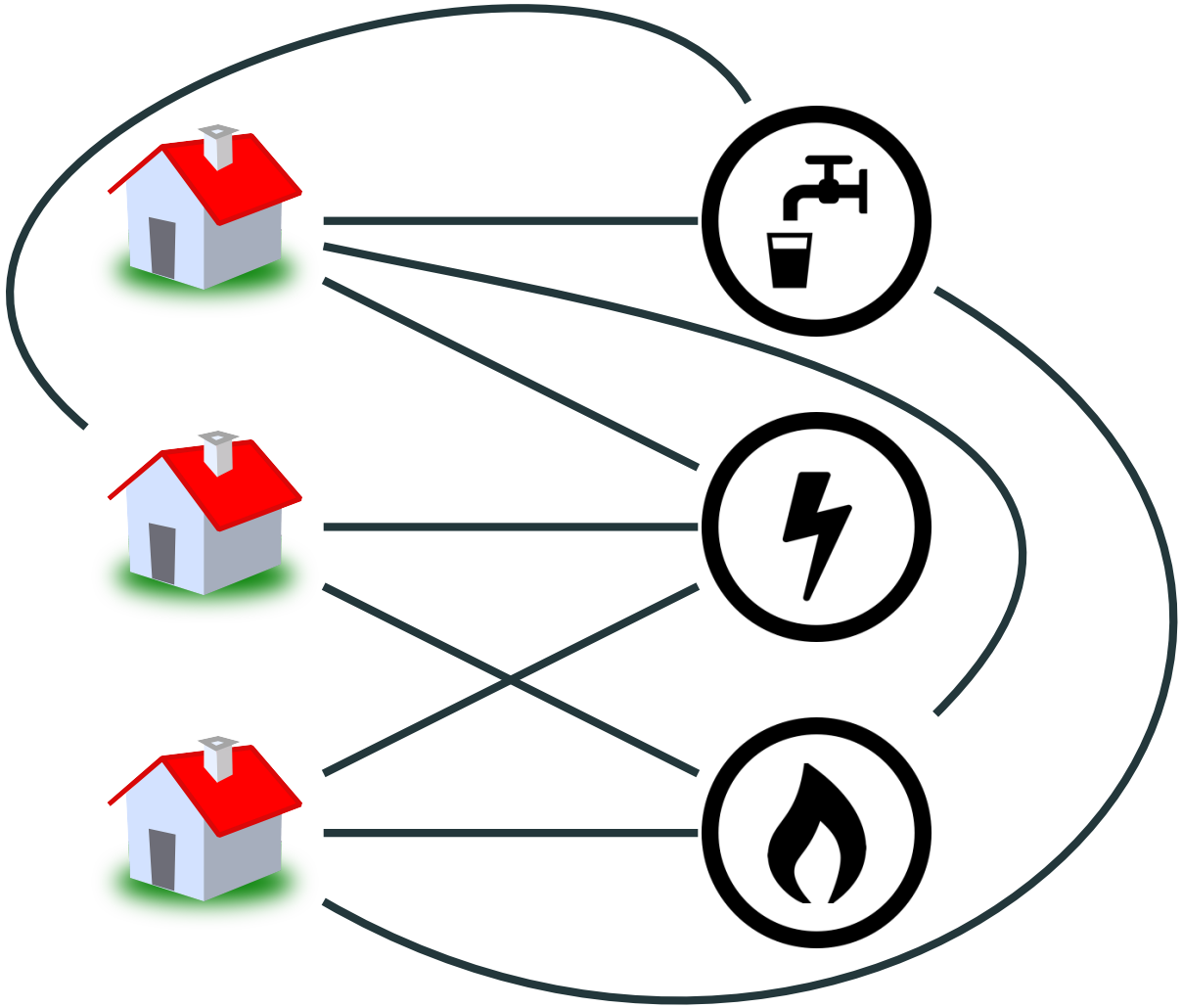




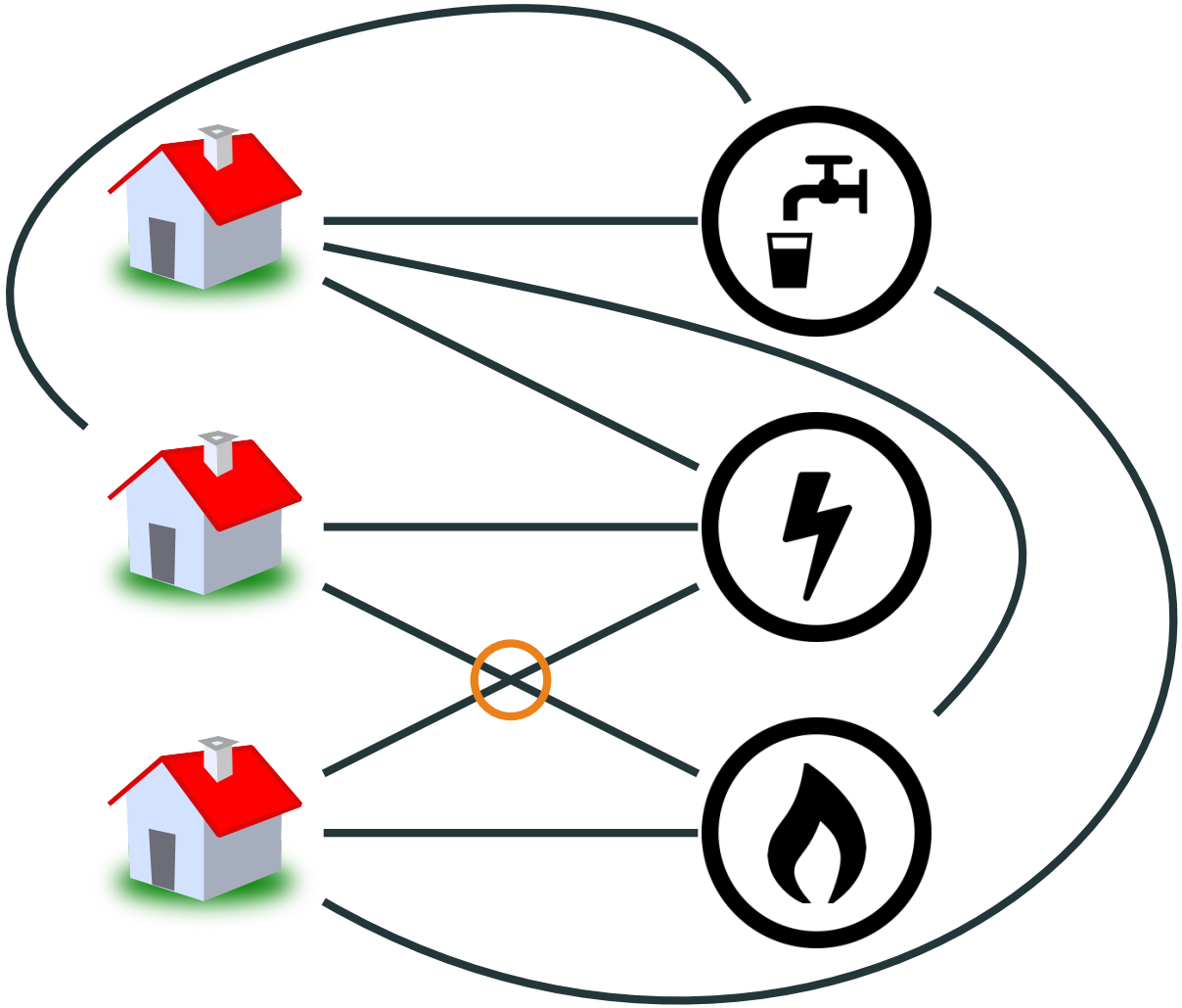




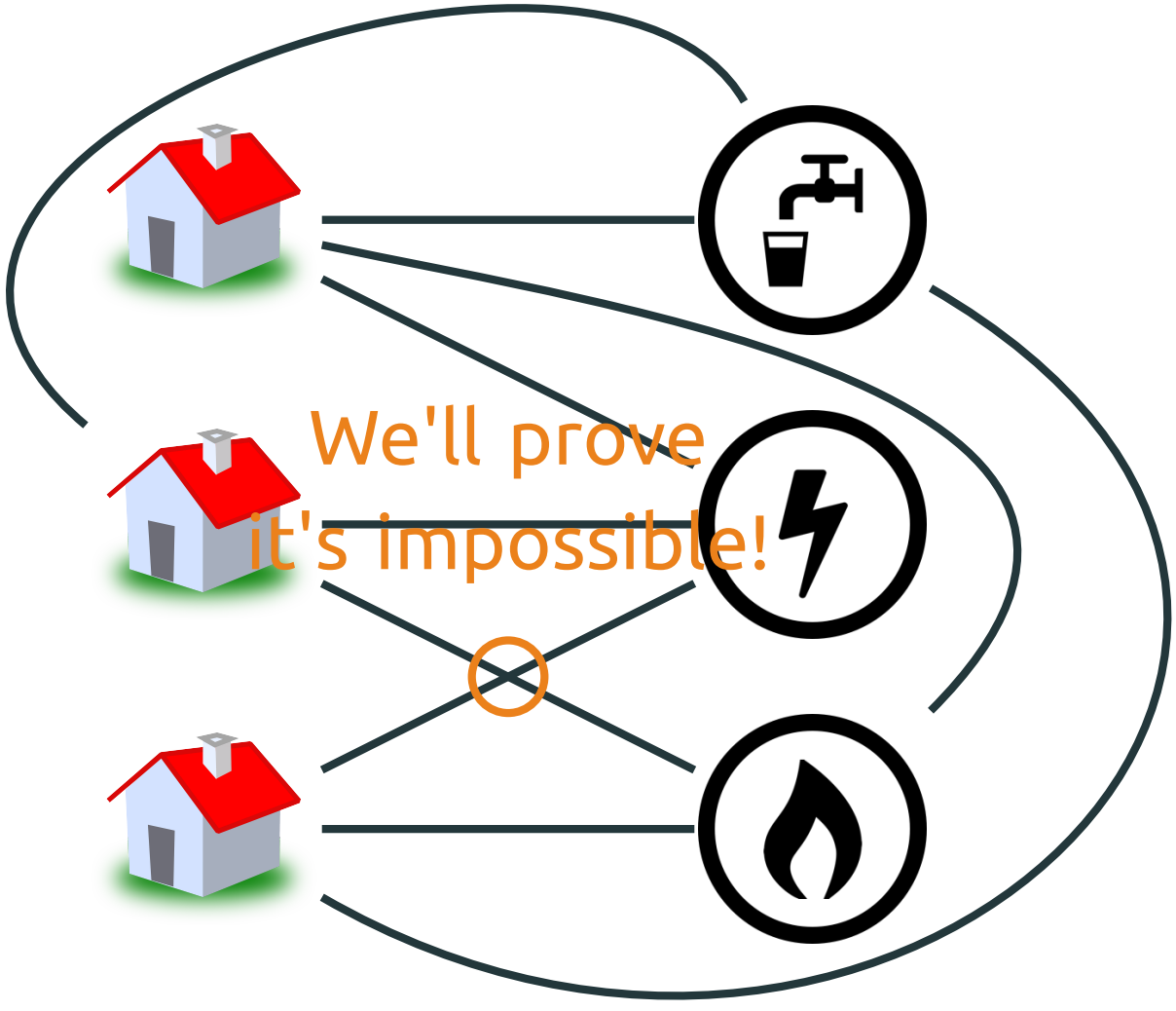
Three Utilities Problem



Three Utilities Problem



Three Utilities Problem



Planar Graphs

- A graph is called Planar if it can be drawn in the plane such that its edges do not meet except at their end points

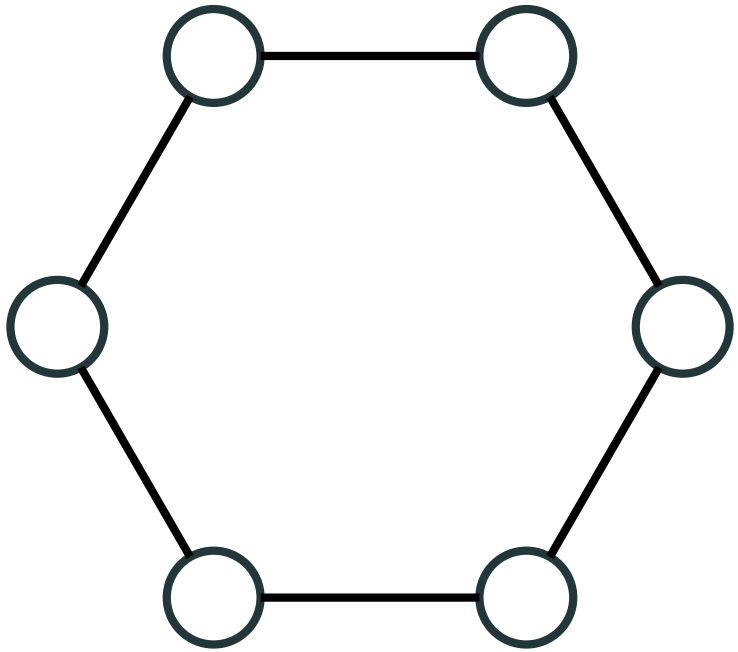
Planar Graphs

- A graph is called Planar if it can be drawn in the plane such that its edges do not meet except at their end points

- Even if you usually draw a graph with intersecting edges, it is Planar if it can be drawn without crossing edges

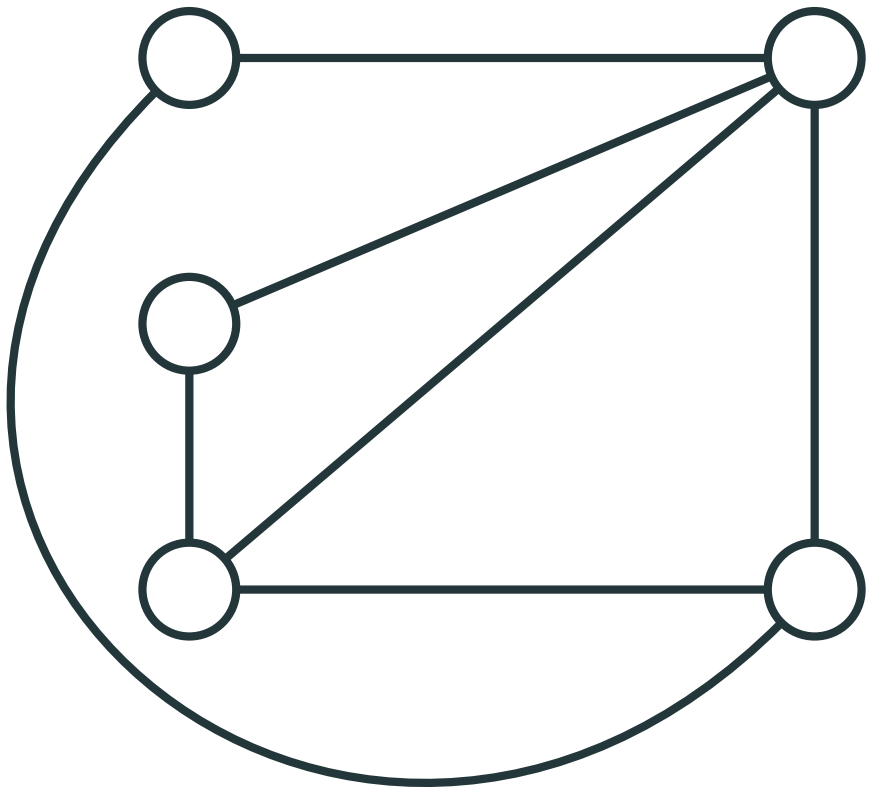
Planar Graphs: Examples

This graph is planar because it can be drawn without crossing edges

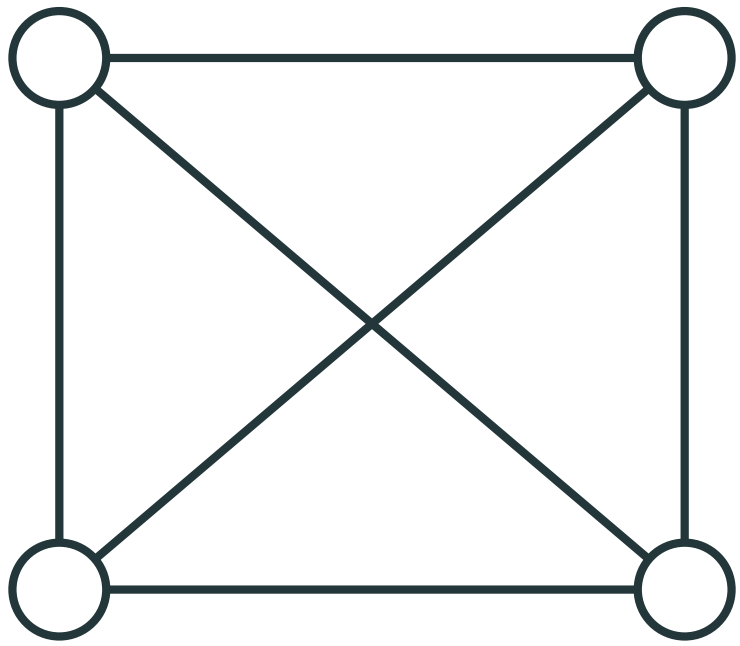


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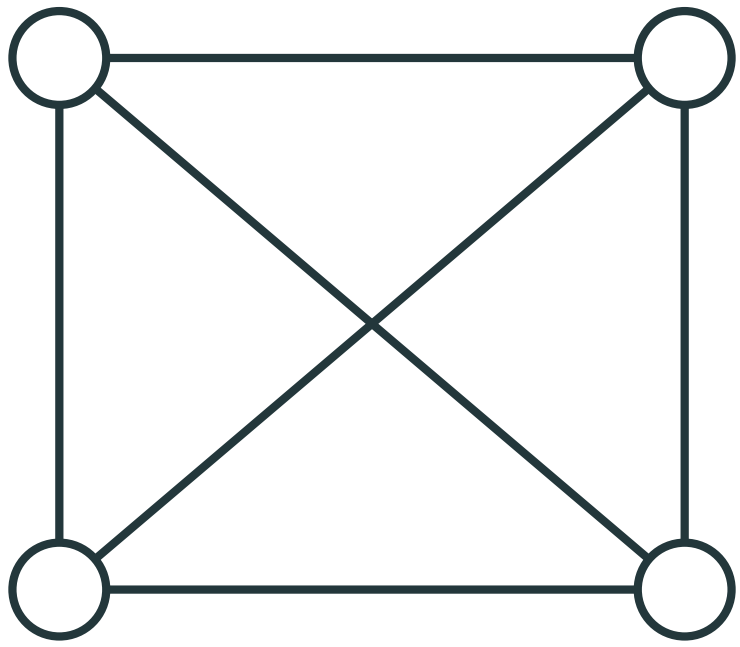


Planar Graphs: Examples



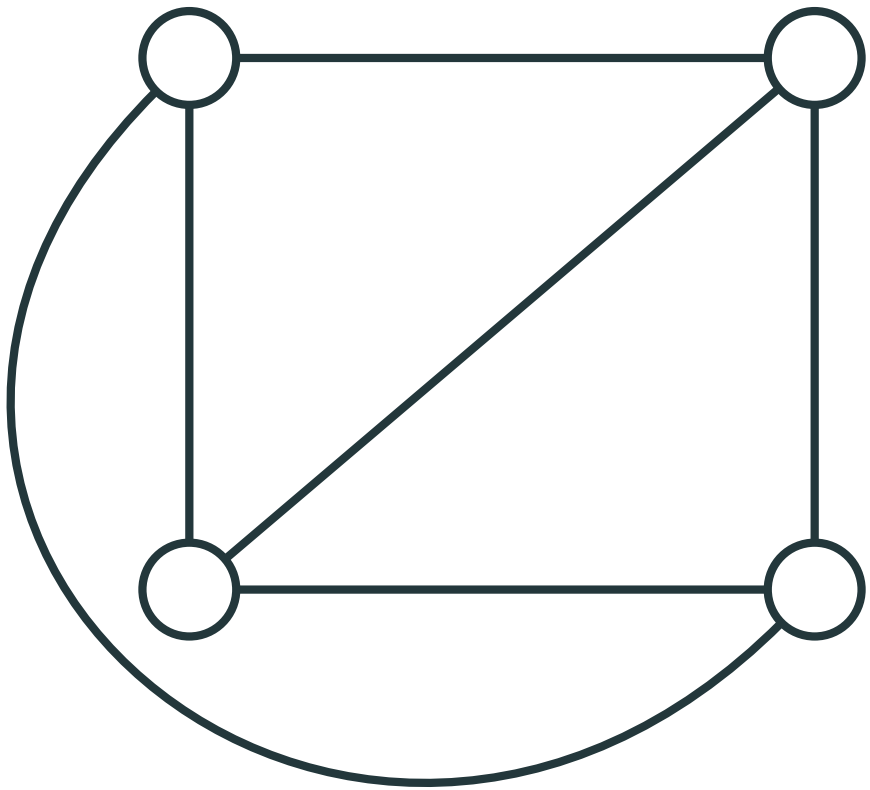
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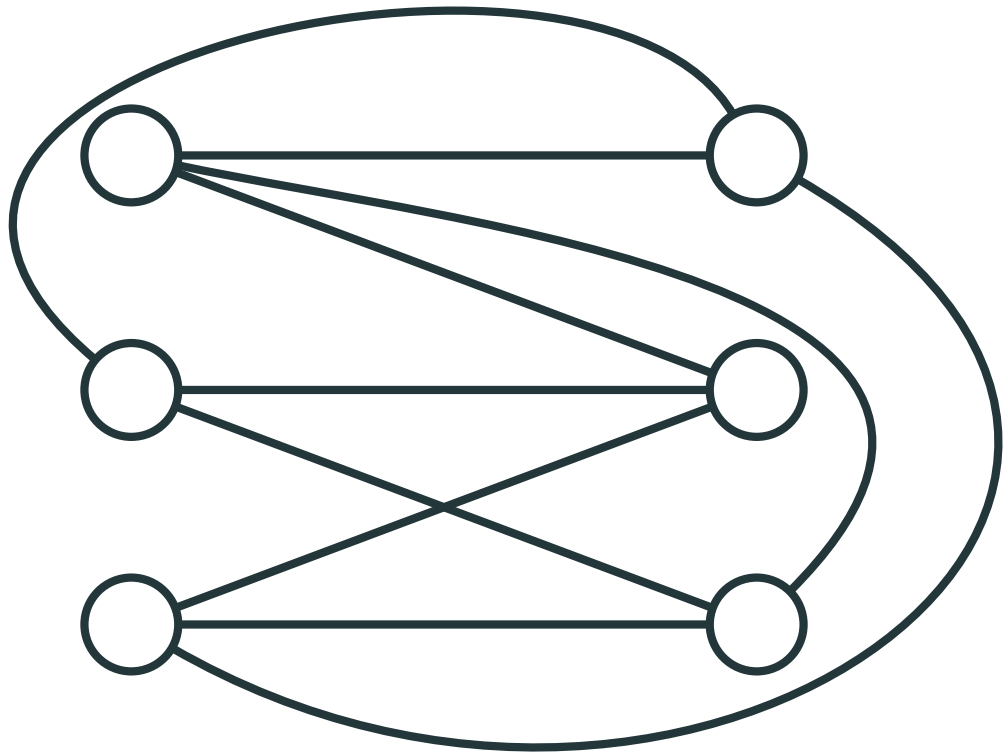


Planar Graphs: Examples

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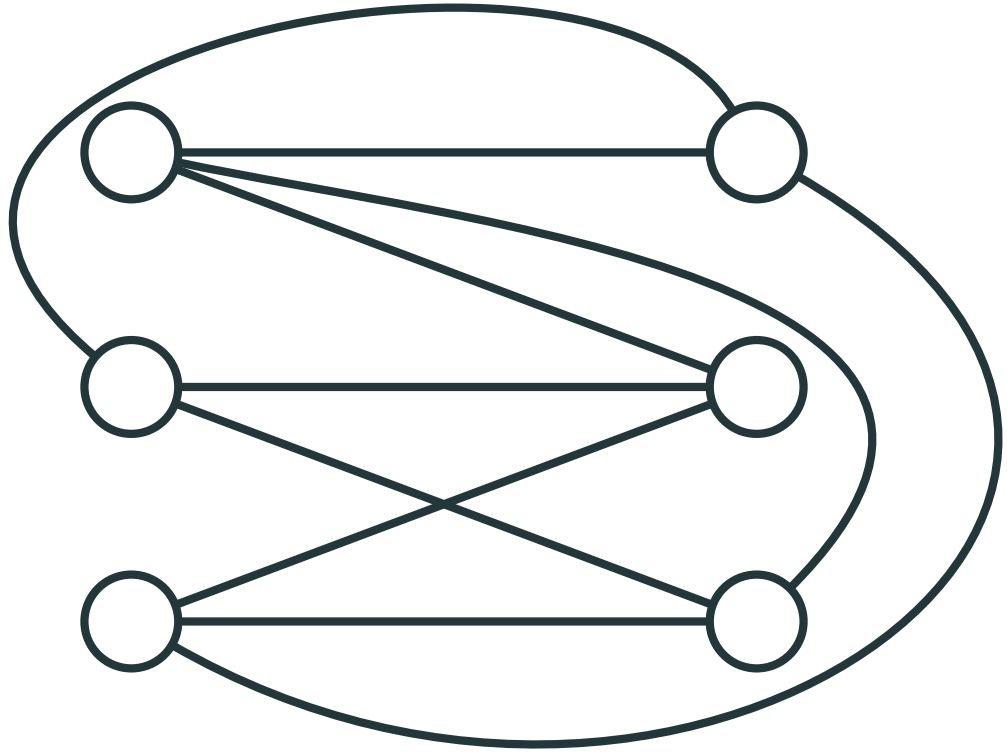


Planar Graphs: Examples



Planar Graphs: Examples

This graph is not planar because it cannot be drawn without crossing edges (we'll prove it later)



Maps and Planar Graphs



Maps and Planar Graphs





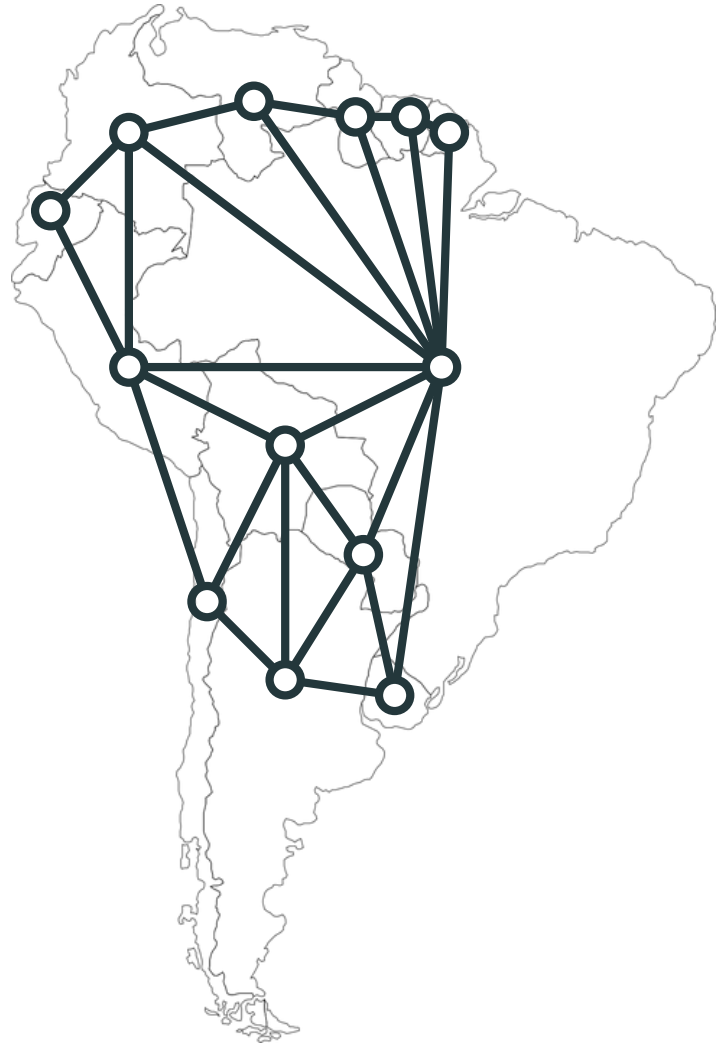
Maps and Planar Graphs





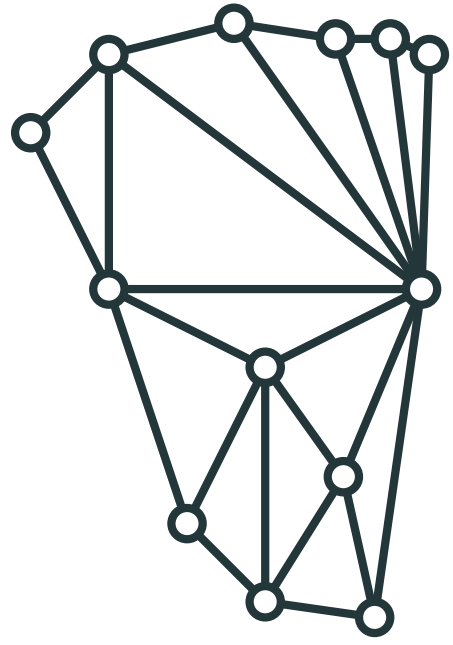
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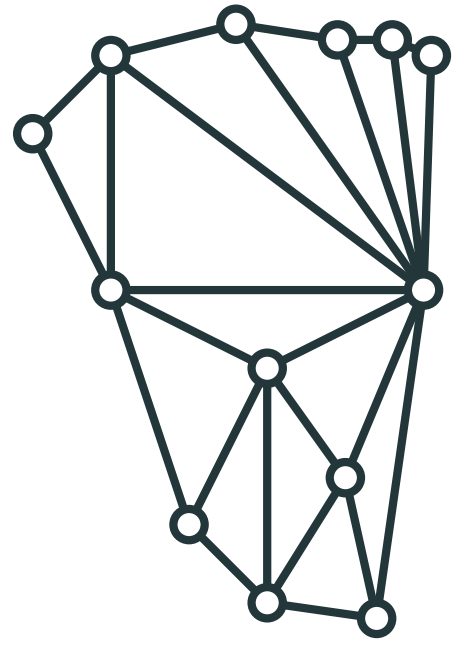


Maps and Planar Graphs

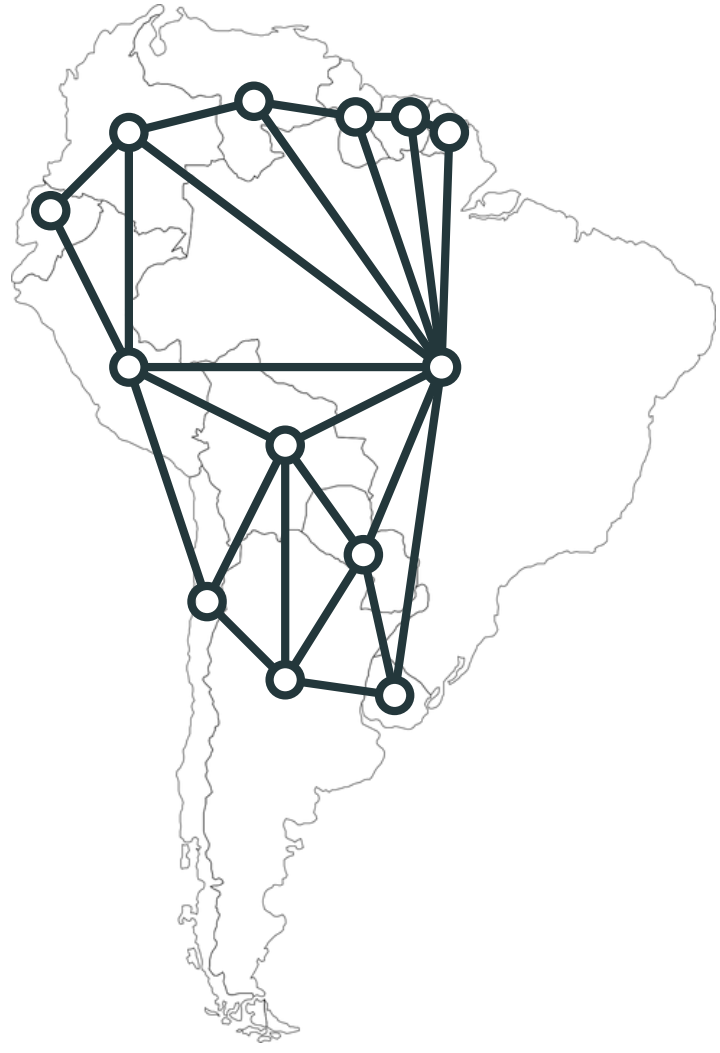




Maps and Planar Graphs

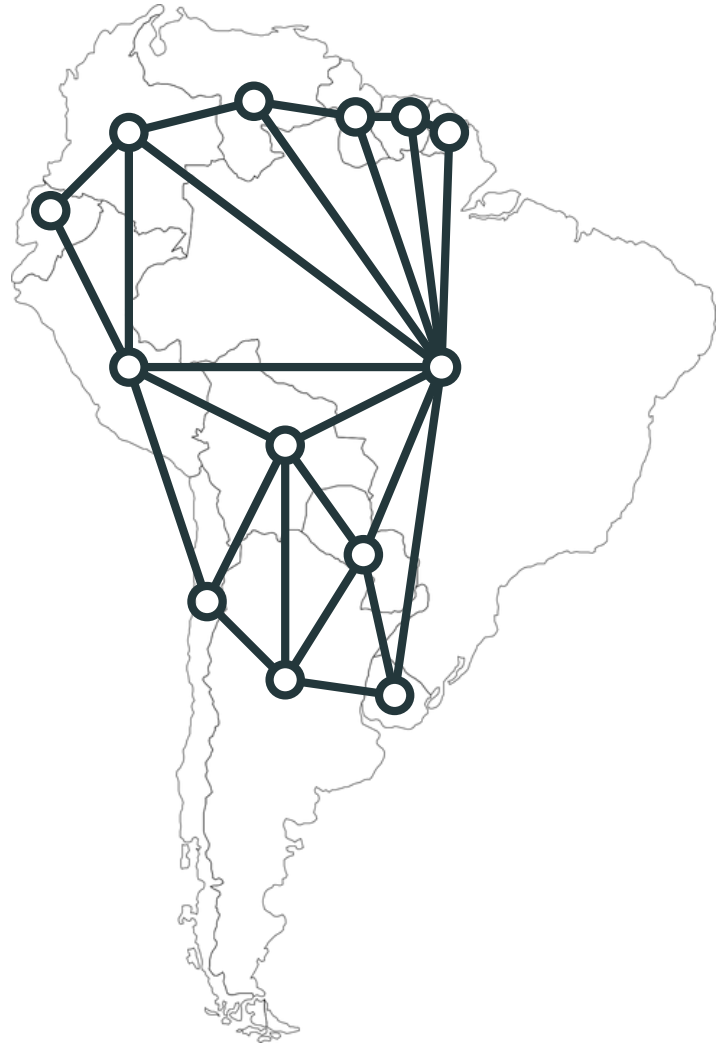


Maps and Planar Graphs



Maps and Planar Graphs





Outline

Subway Lines

Planar Graphs

Euler's Formula

Applications of Euler's Formula

Graph Faces

- Let us fix some Drawing of a planar graph

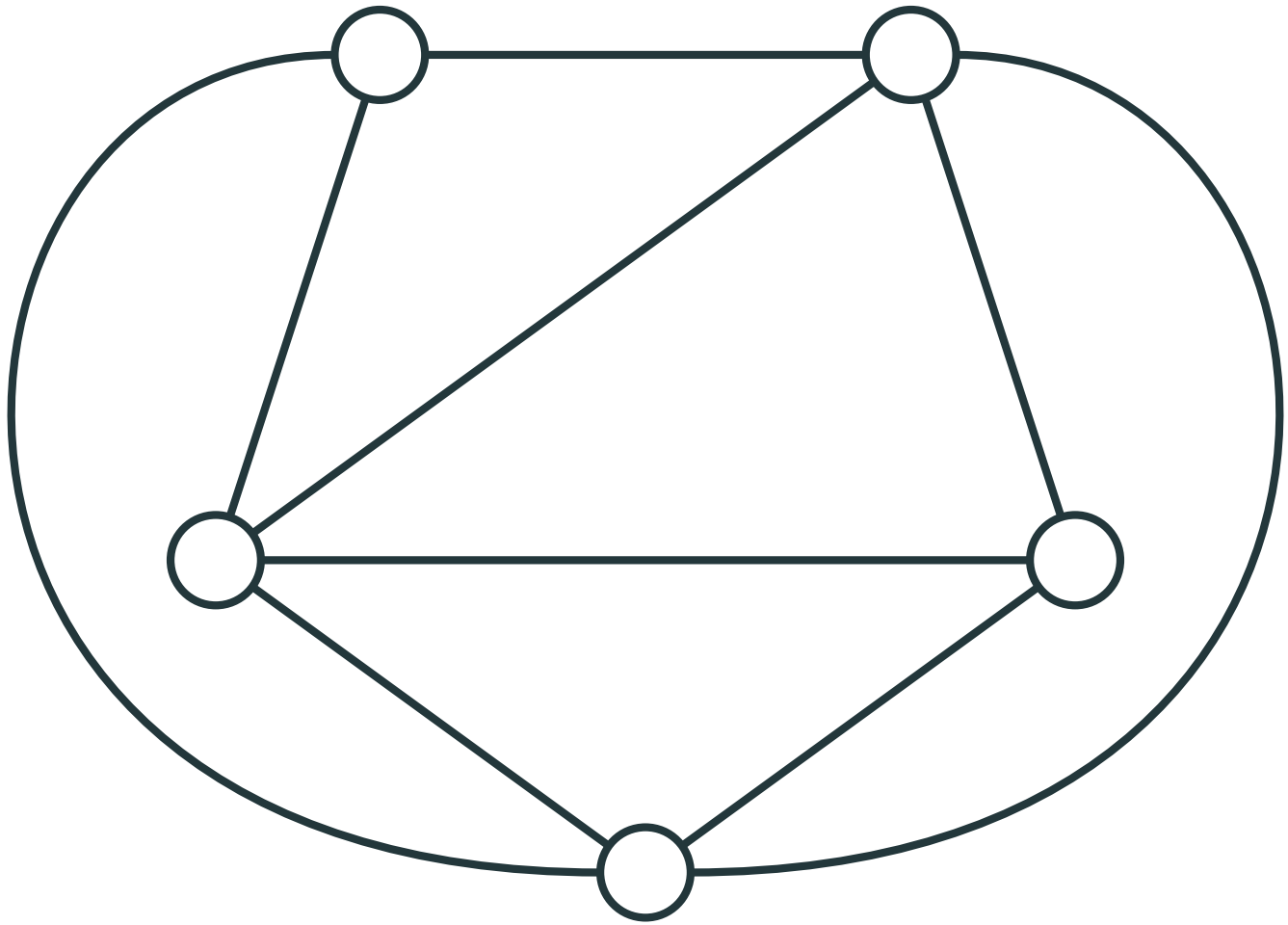
Graph Faces

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- Then a Face of this graph is a region bounded by the edges of the graph

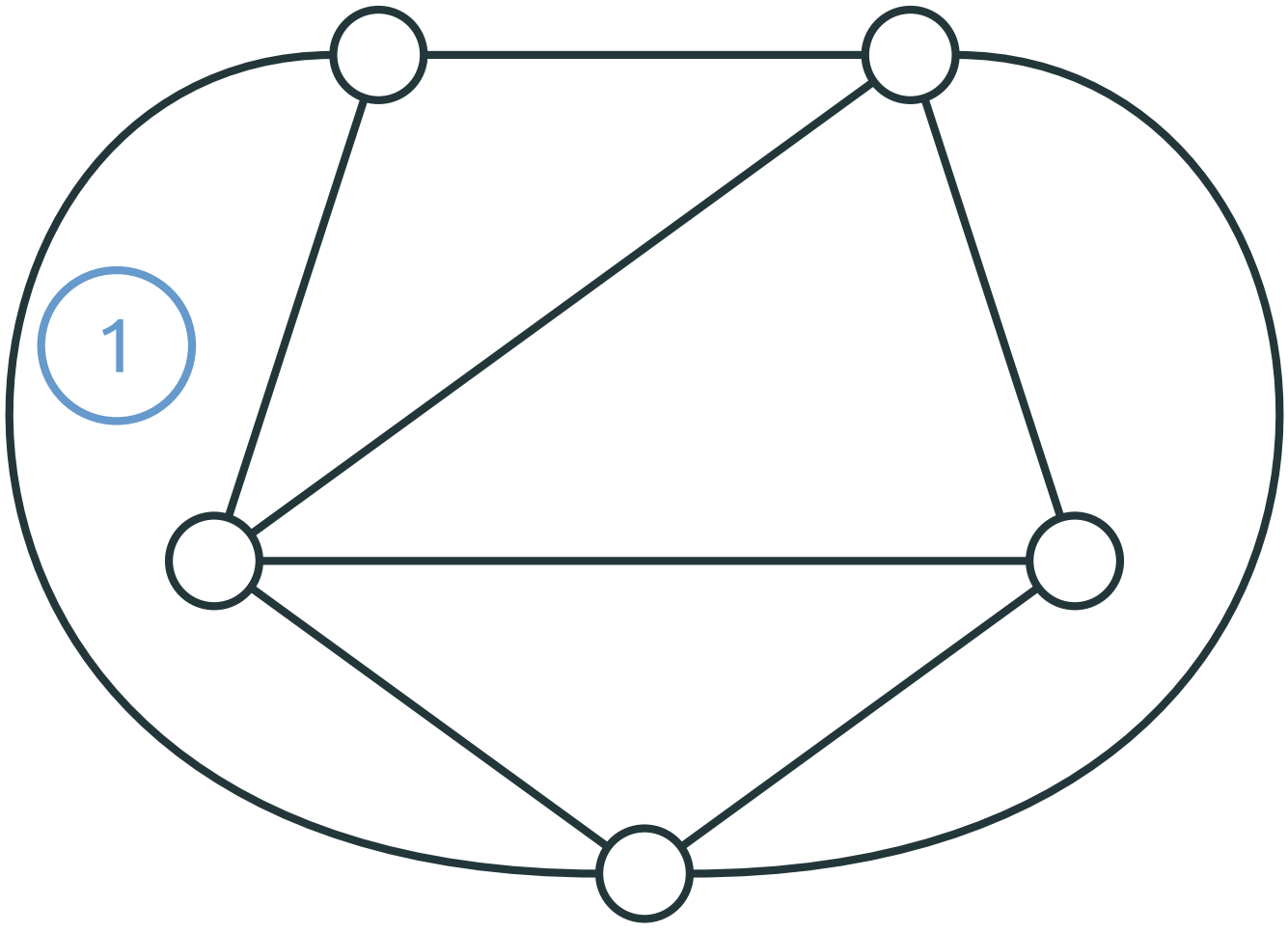
Graph Faces

- Let us fix some Drawing of a planar graph
- Then a Face of this graph is a region bounded by the edges of the graph
- Note that there is one infinitely large outer face

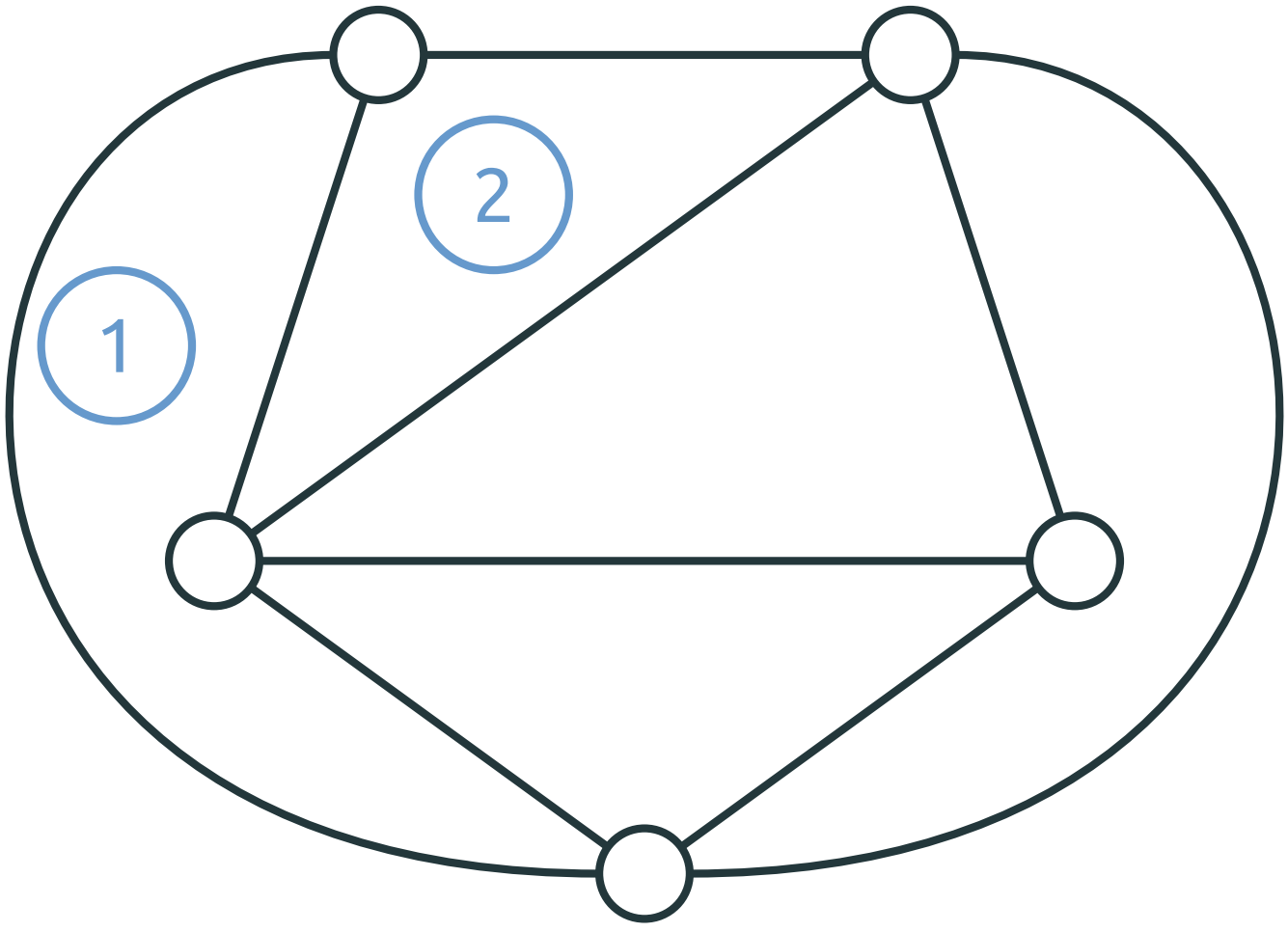
Graph Faces: Examples



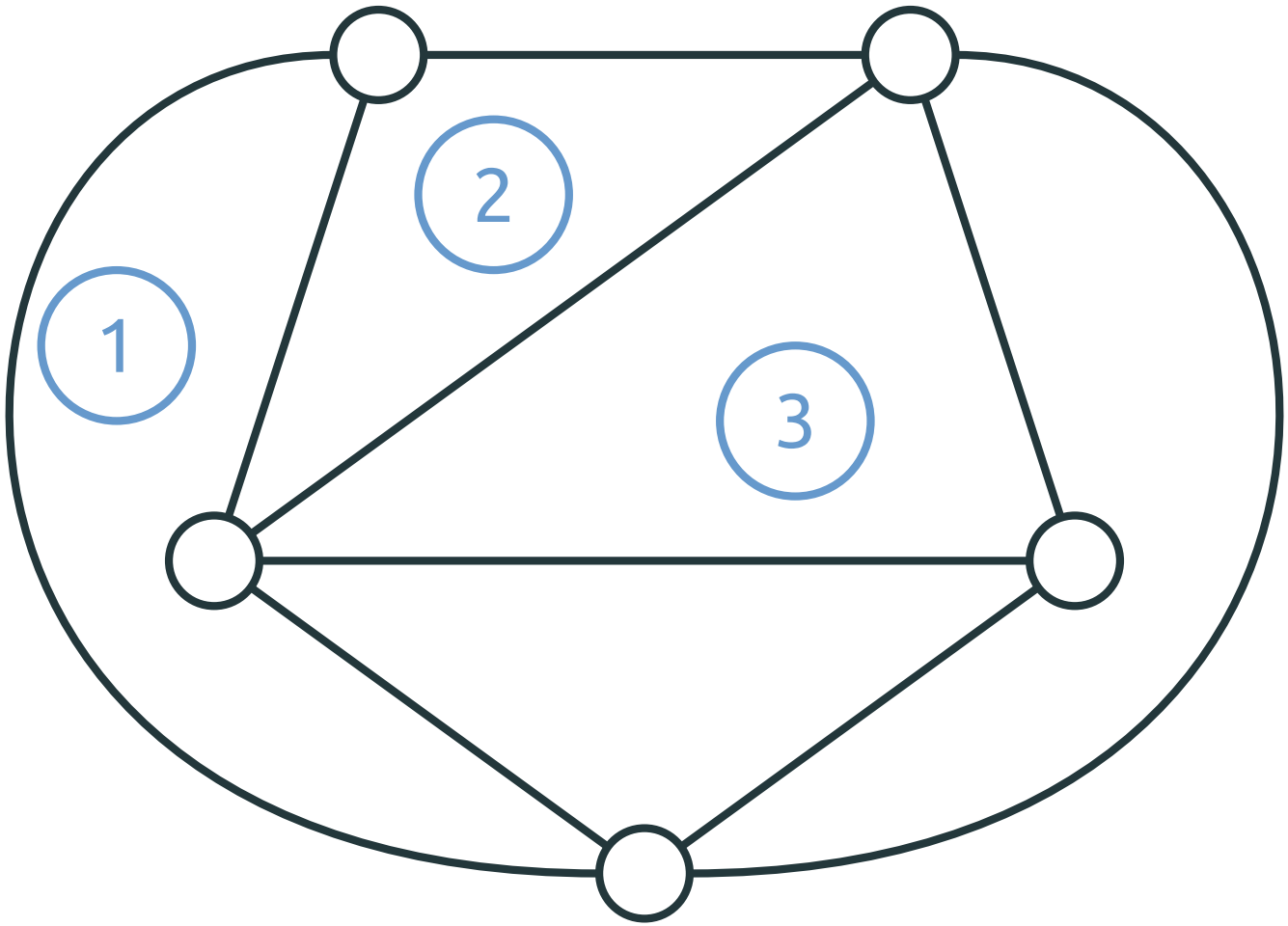
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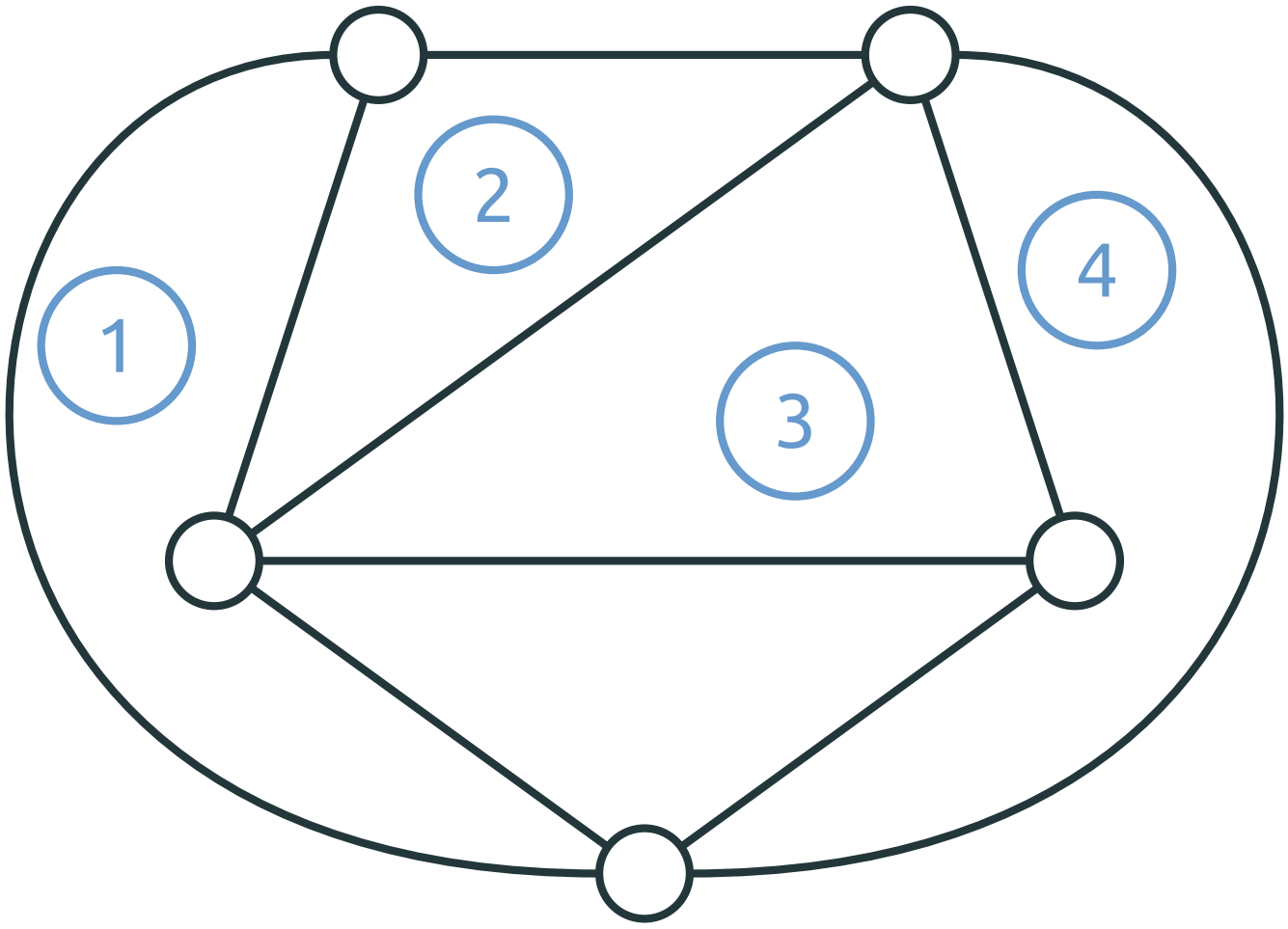
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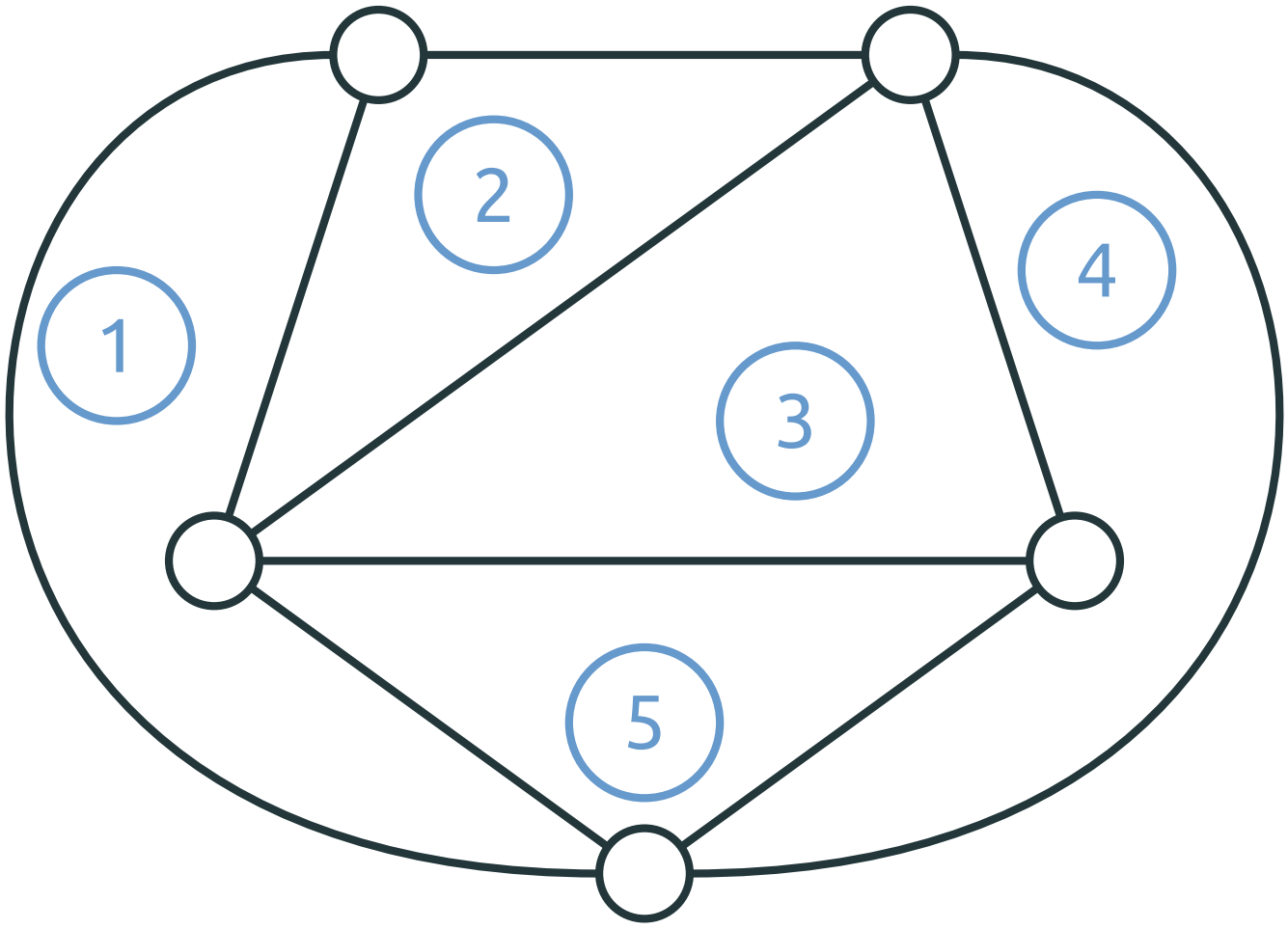
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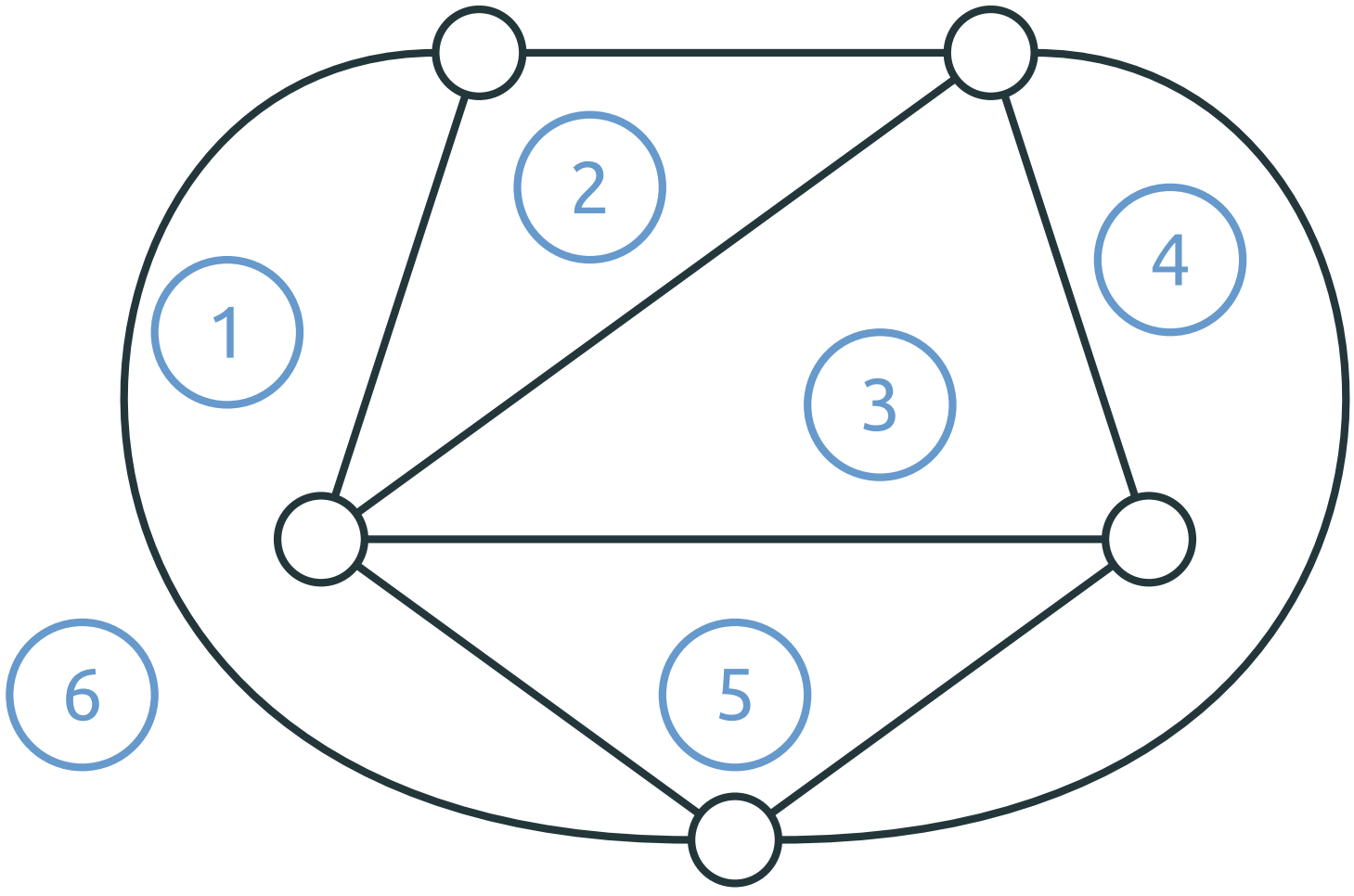
Graph Faces: Examples



Graph Faces: Examples



Graph Faces: Examples



Euler's Formula

Theorem

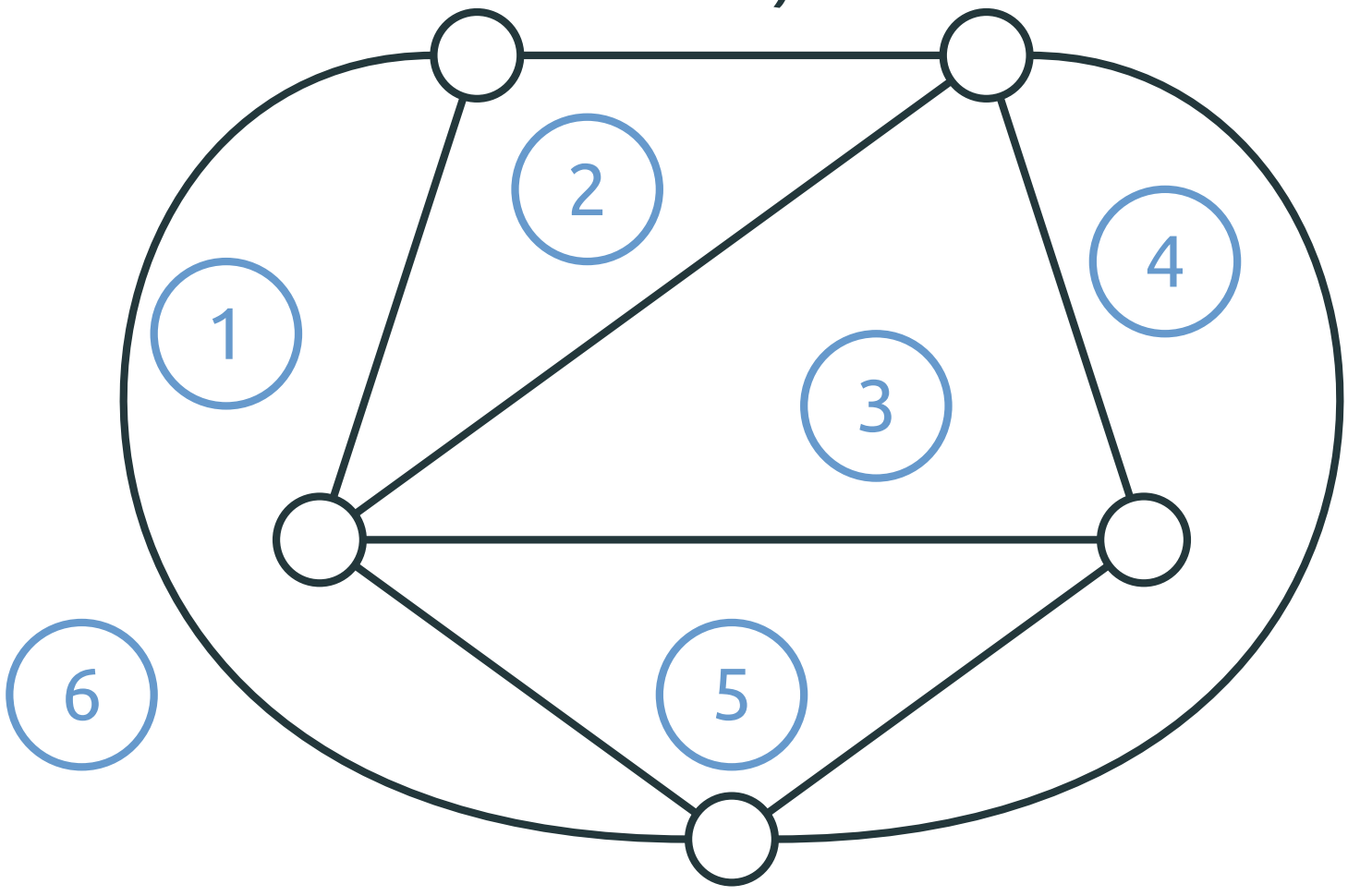
Let G be a connected planar graph drawn in the plane without edge intersections. Then

$$v - e + f = 2 \ ,$$

where v is the number of vertices, e is the number of edges, f is the number of faces in this drawing of G.

Euler's Formula

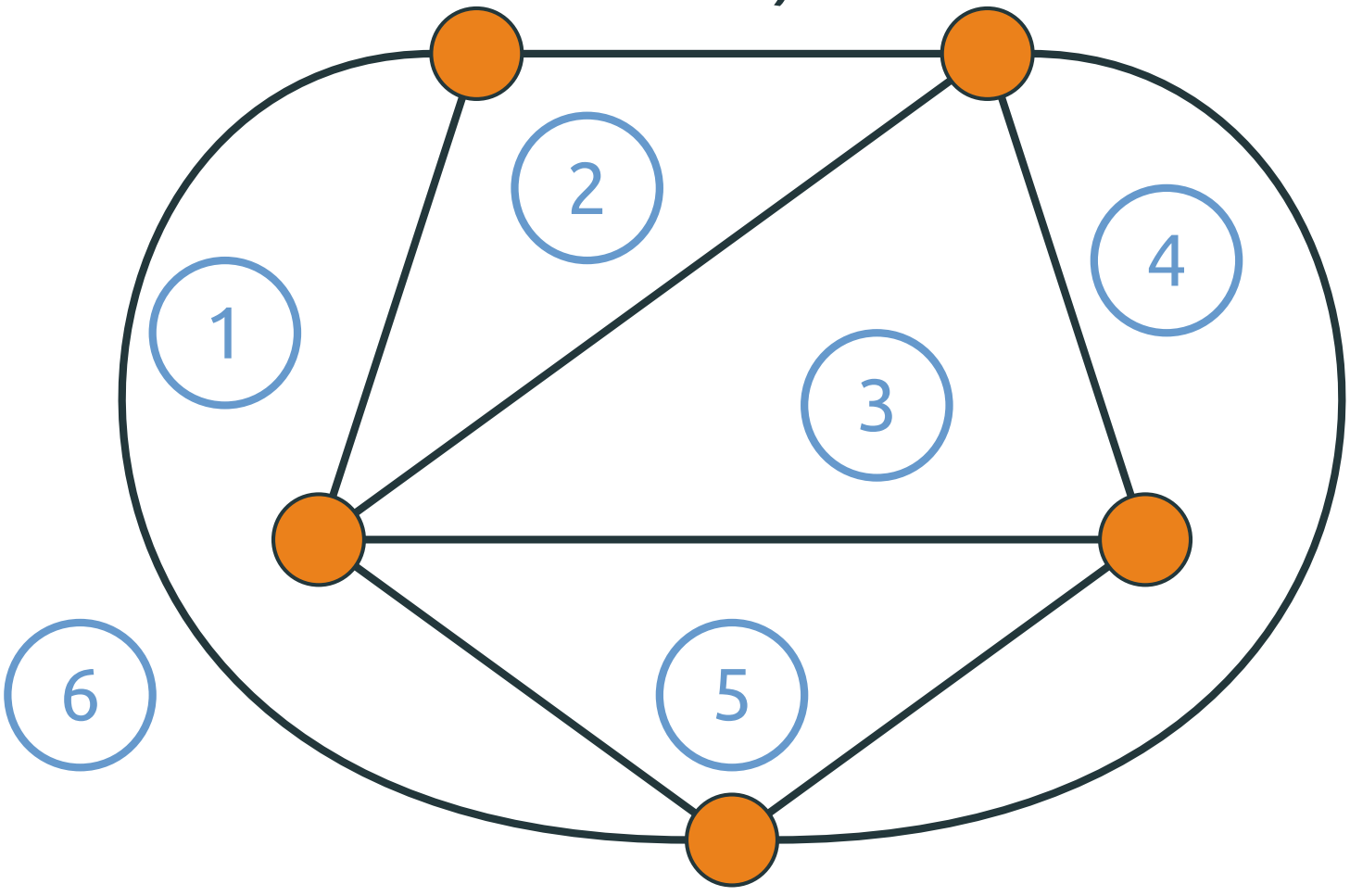
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Euler's Formula

5 –

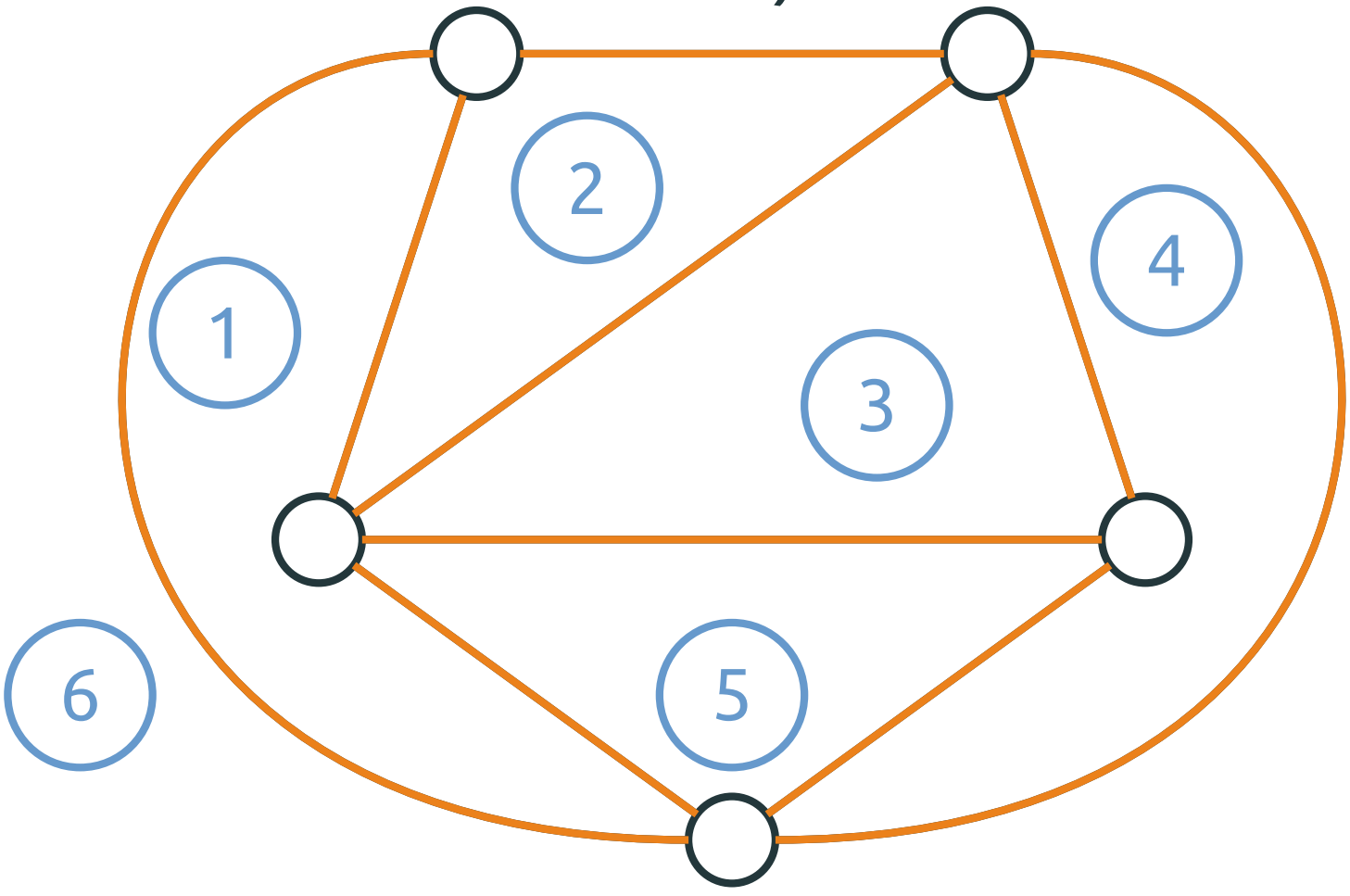
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Euler's Formula

5− 9+

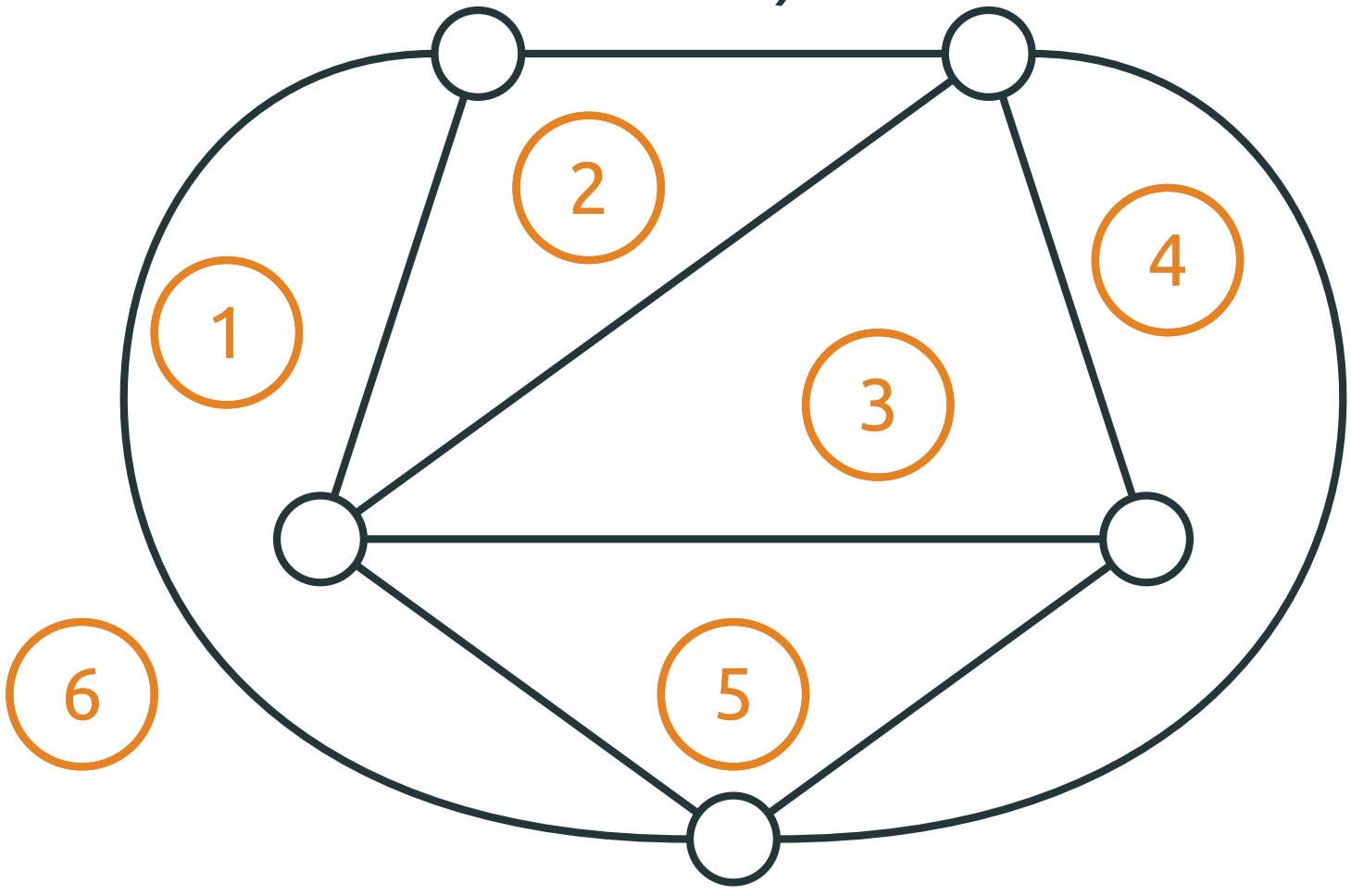
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Euler's Formula

$5 - 9 + 6$

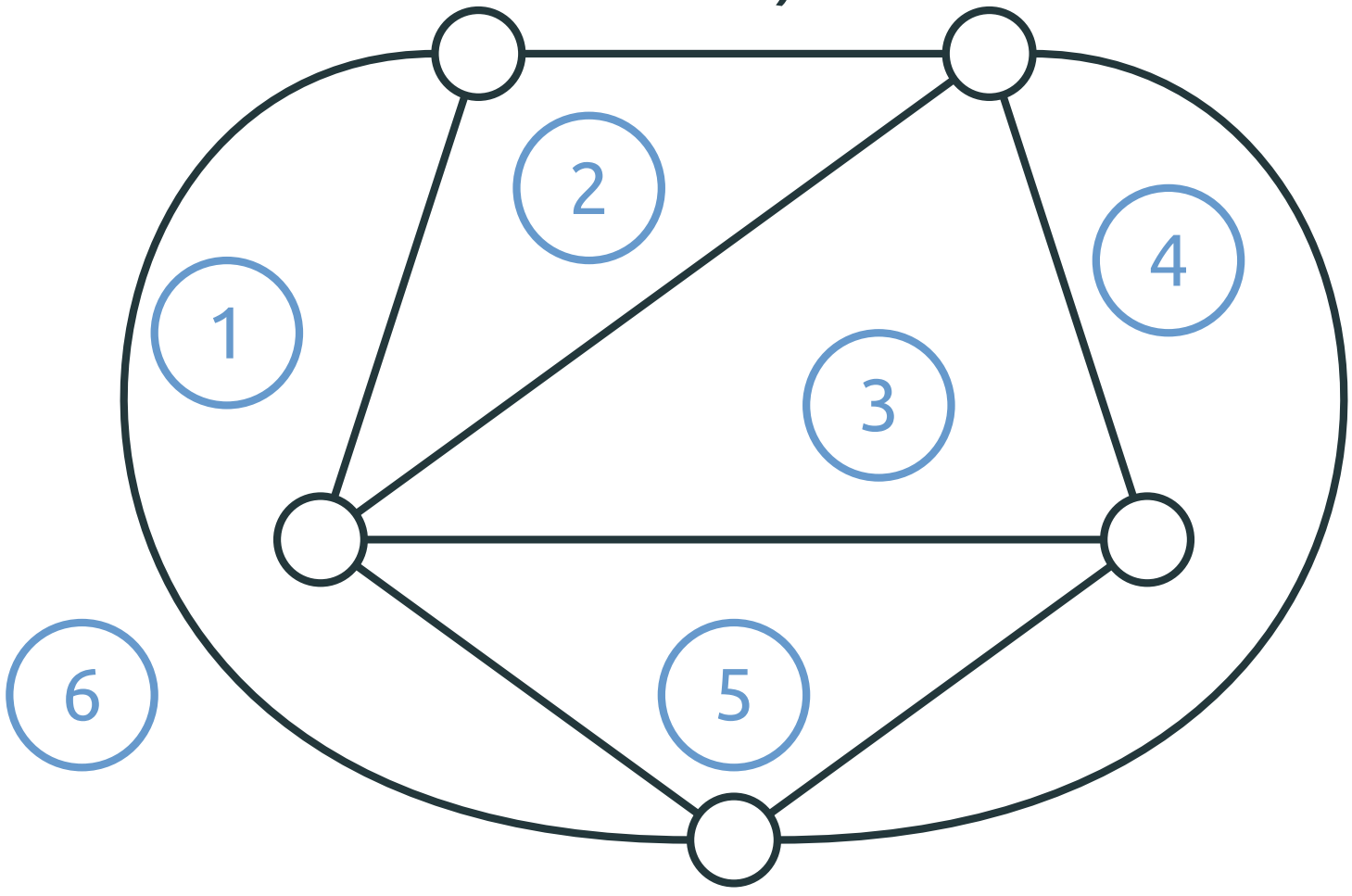
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Euler's Formula

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Euler's Formula: Proof

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- Induction Hypothesis. The formula holds for all graphs with $\leq c$ cycles
- Induction Step. We'll prove the formula for G with $c + 1$ cycles, v vertices, e edges, and f faces

Euler's Formula: Proof

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- The new graph G_1 has $\leq c$ cycles, $f_1 = f - 1$ faces, $e_1 = e - 1$ edges, and $v_1 = v$ vertices

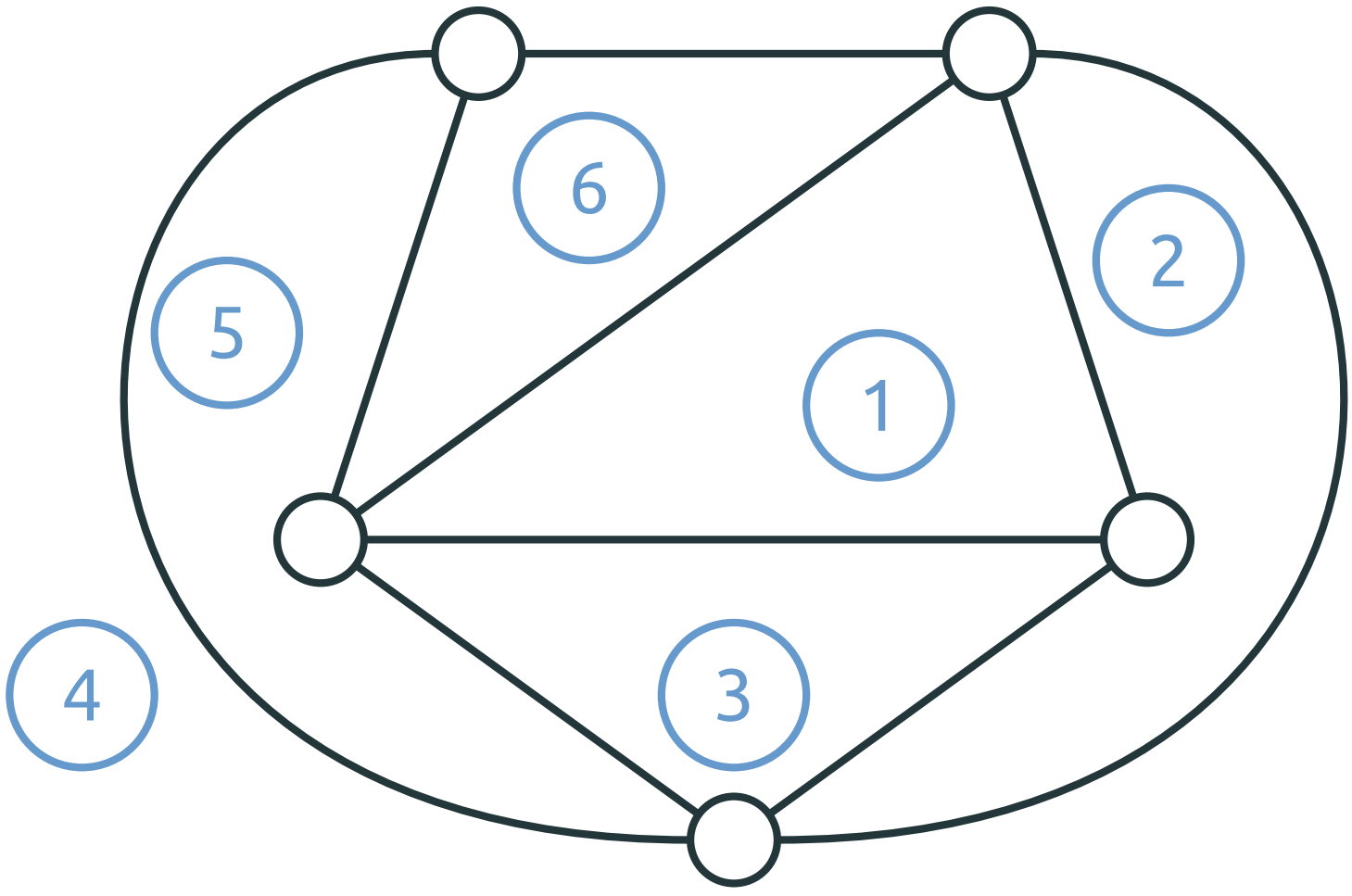
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- By the Induction Hypothesis, $v_1 - e_1 + f_1 = 2$

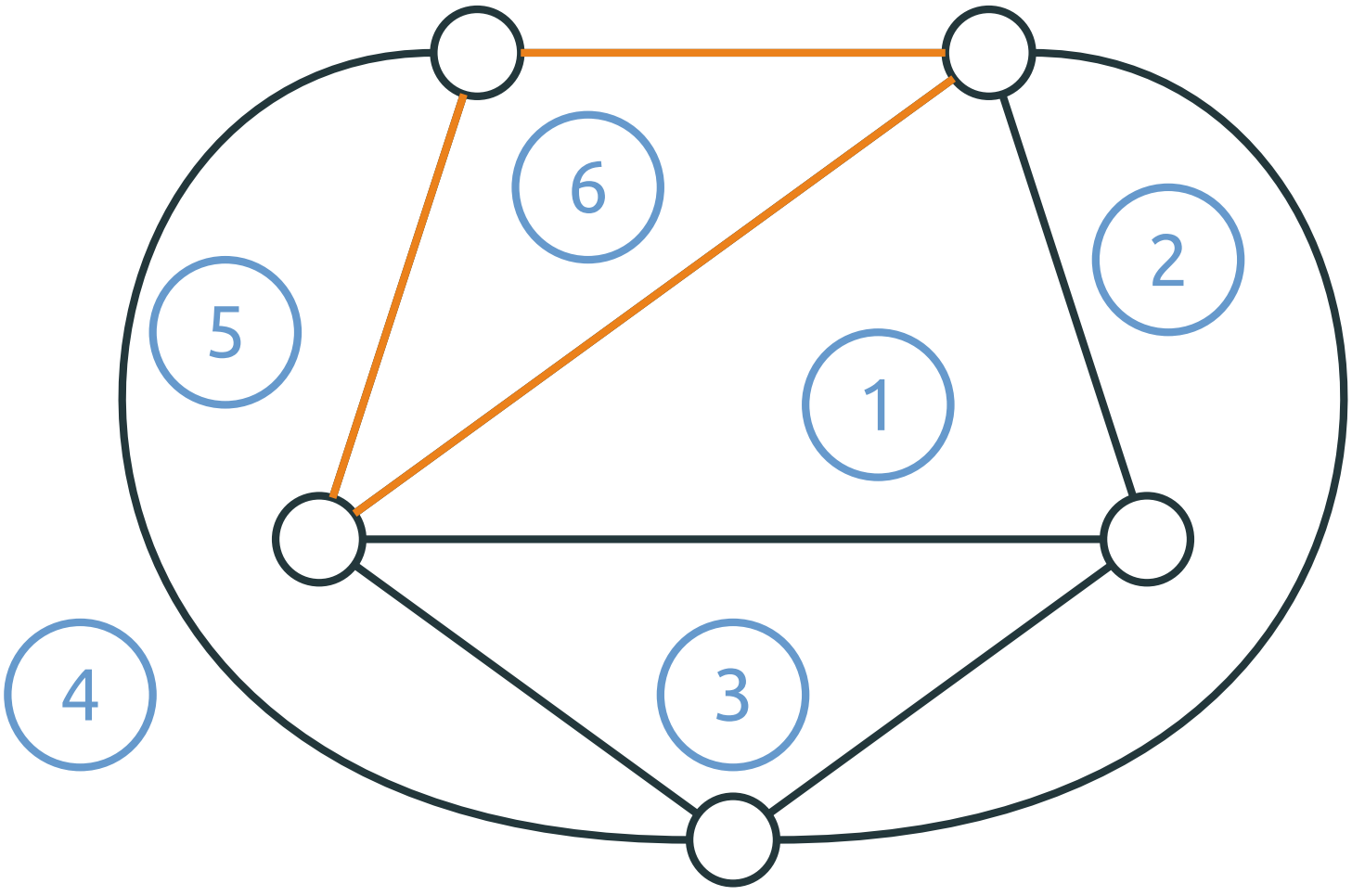
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- By the Induction Hypothesis, $v_1 - e_1 + f_1 = 2$
- Then
$$v - e + f = v_1 - (e_1 + 1) + (f_1 + 1) = v_1 - e_1 + f_1 = 2$$

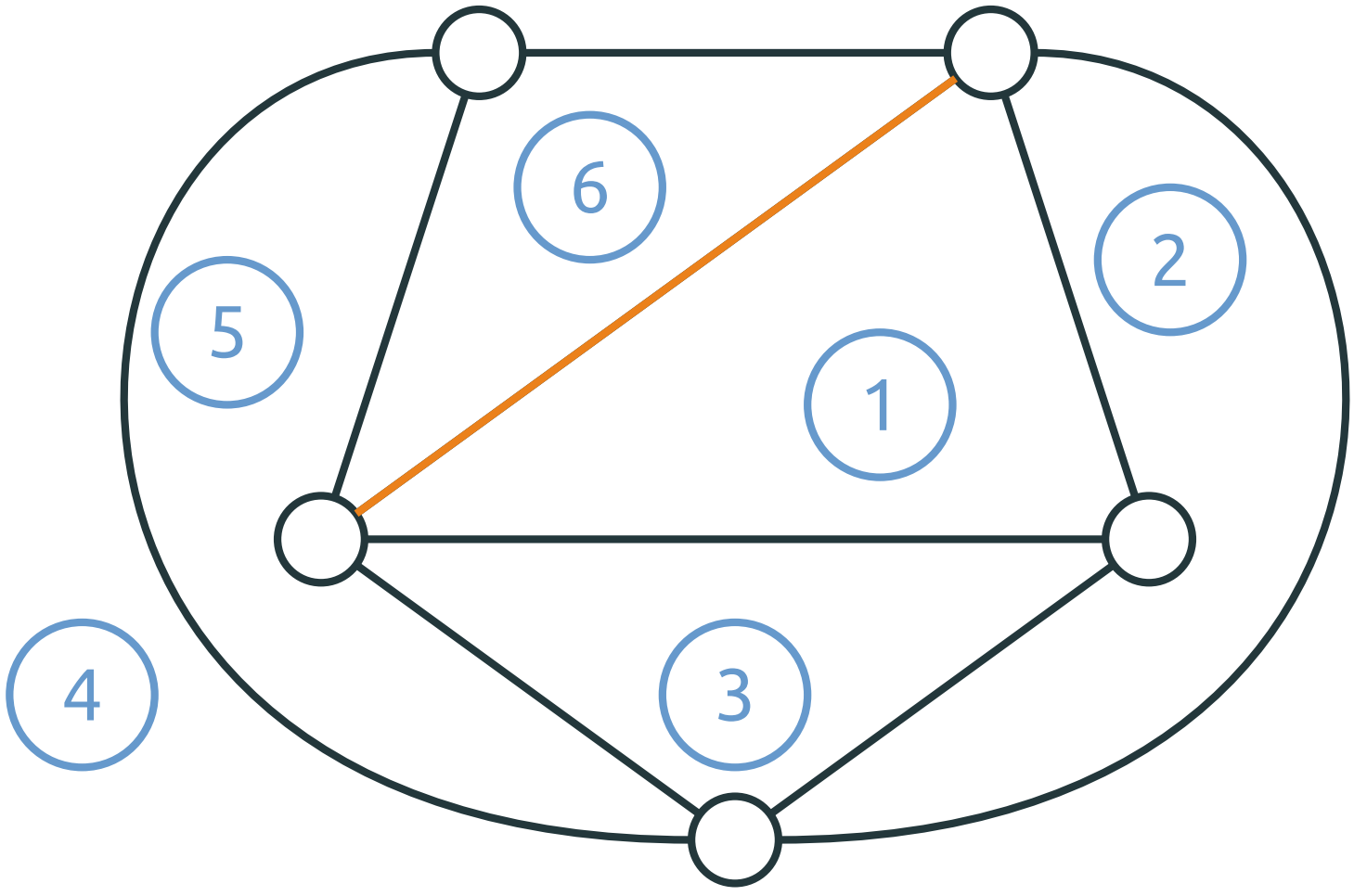
Euler's Formula: Proof



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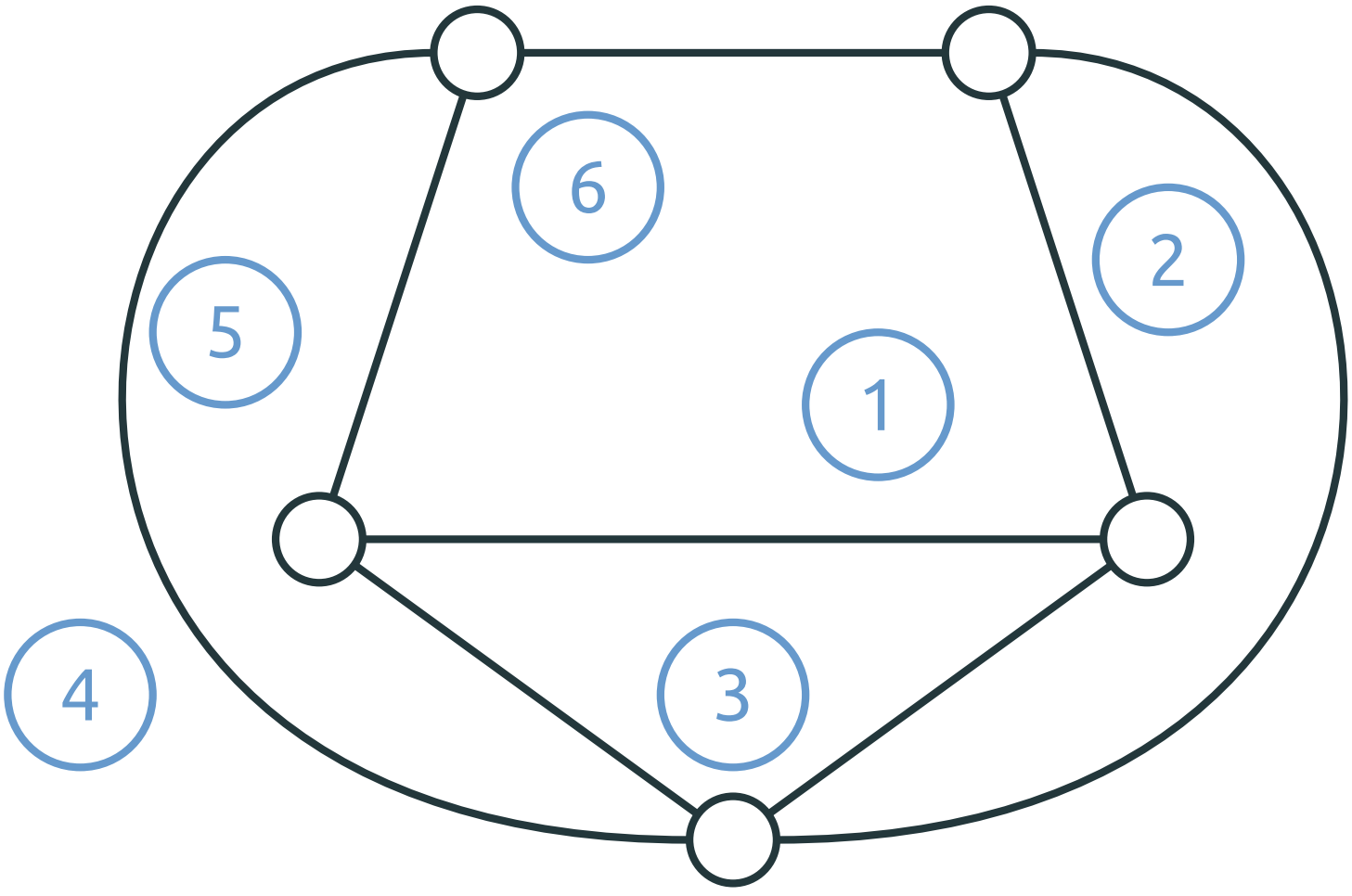


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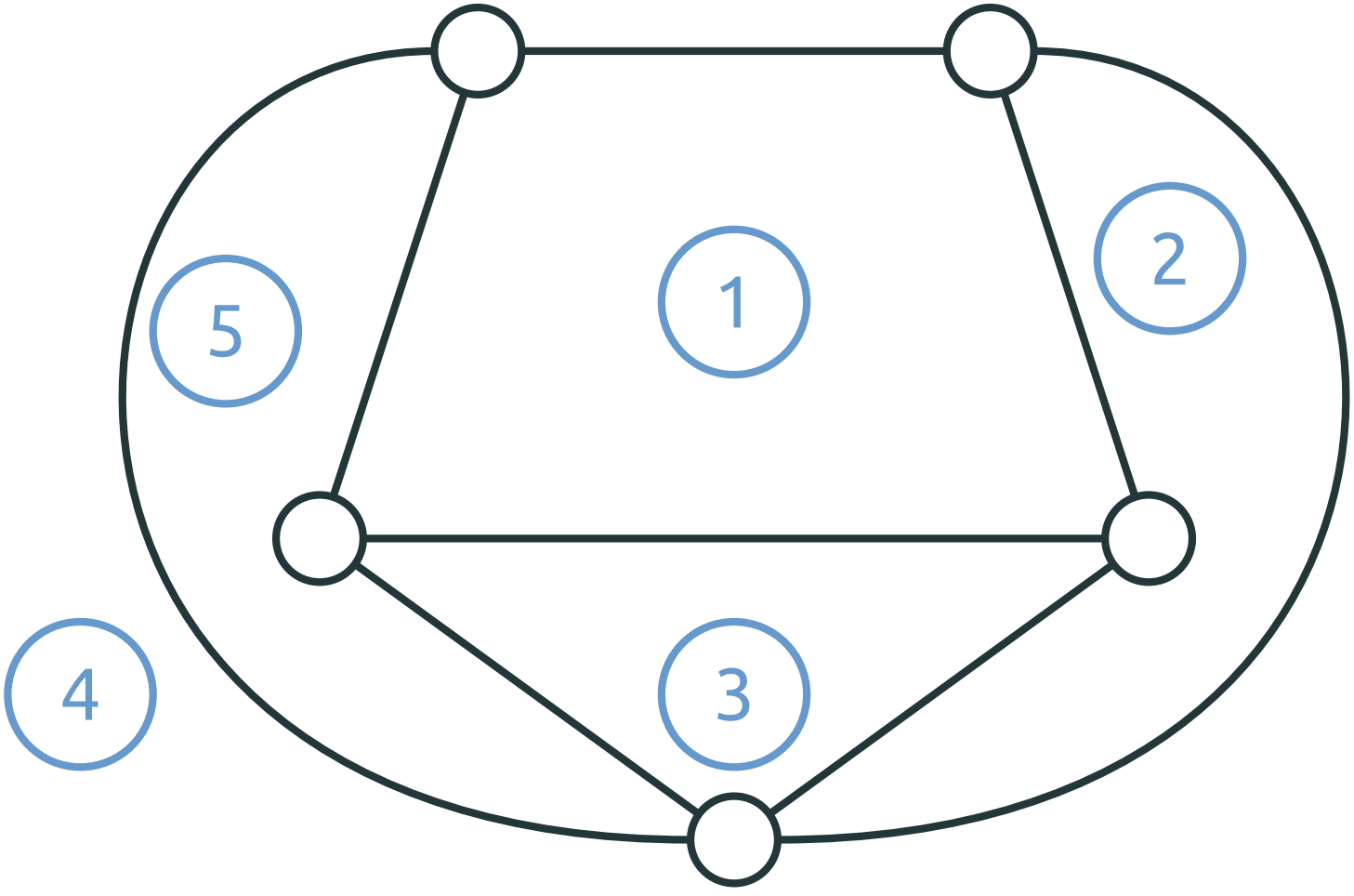
Euler's Formula: Proof

$e_1 = e - 1, v_1 = v$



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Outline

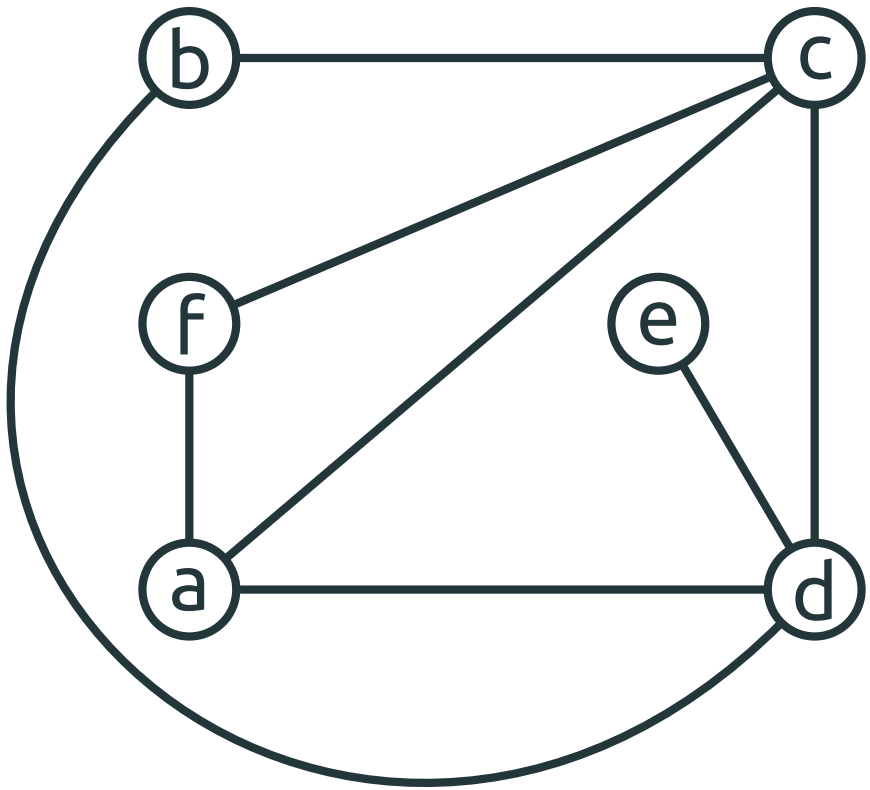
Subway Lines

Planar Graphs

Euler's Formula

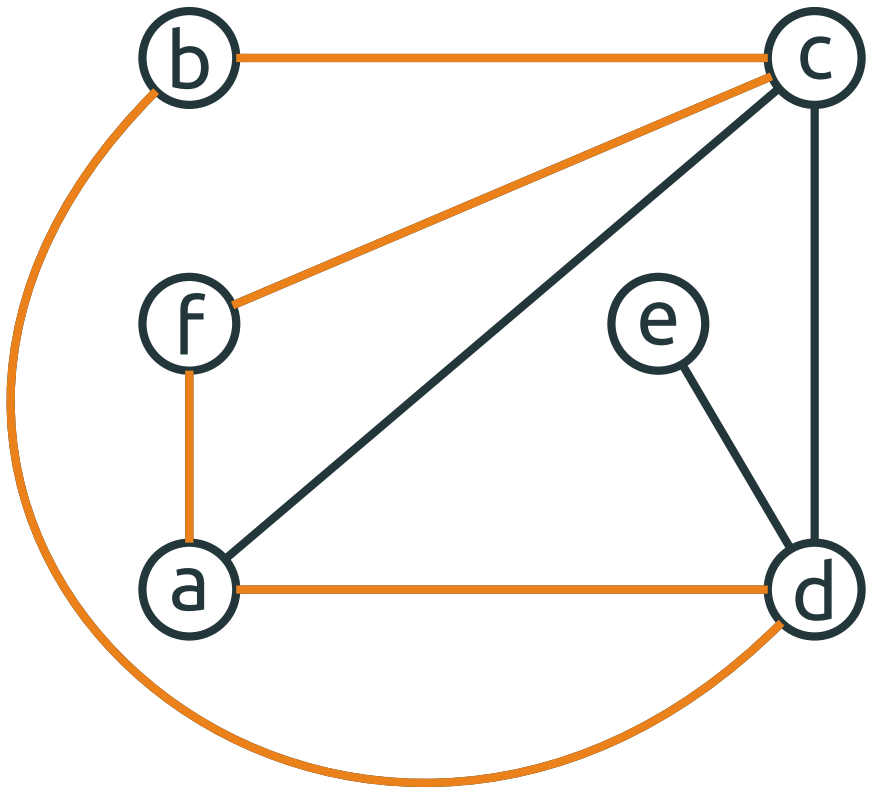
Applications of Euler's Formula

Faces and Edges



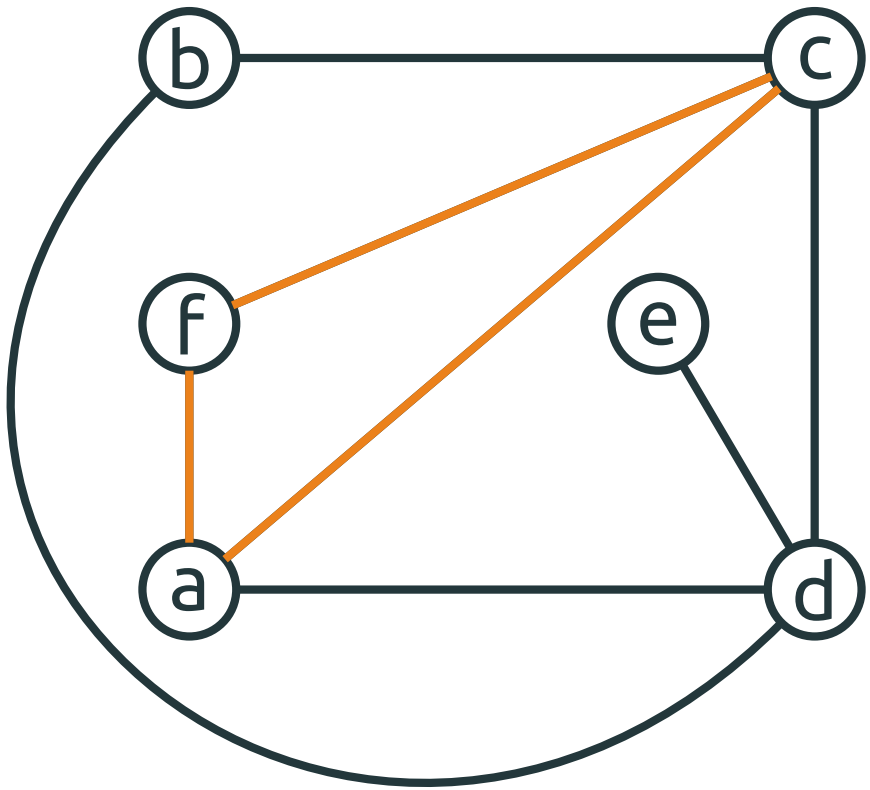
Faces and Edges

This face has 5 edges



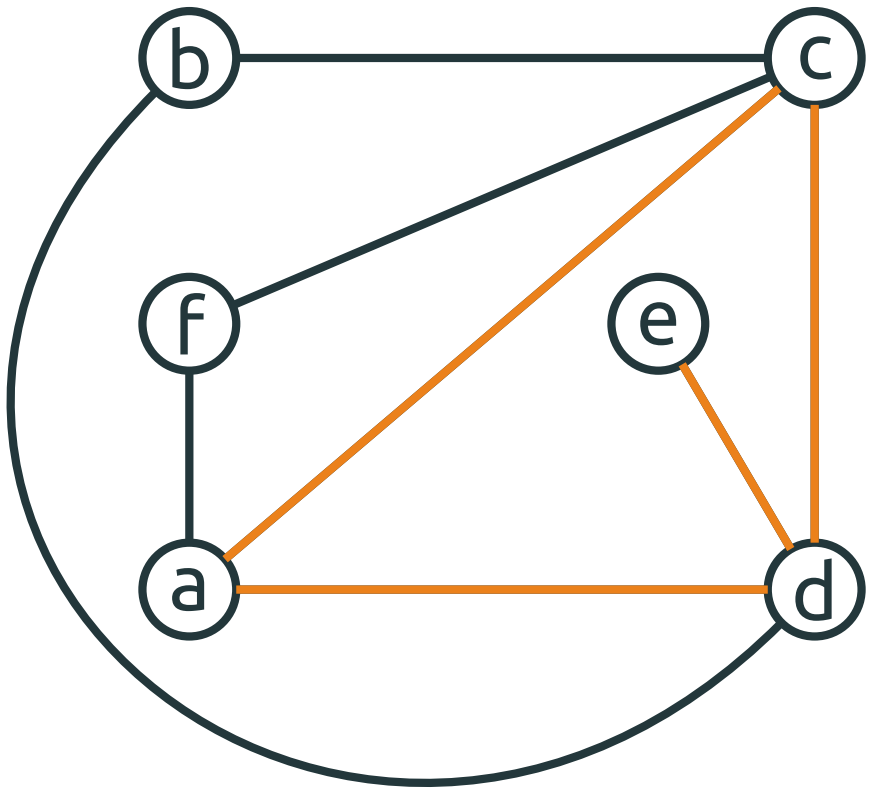
Faces and Edges

This face has 3 edges



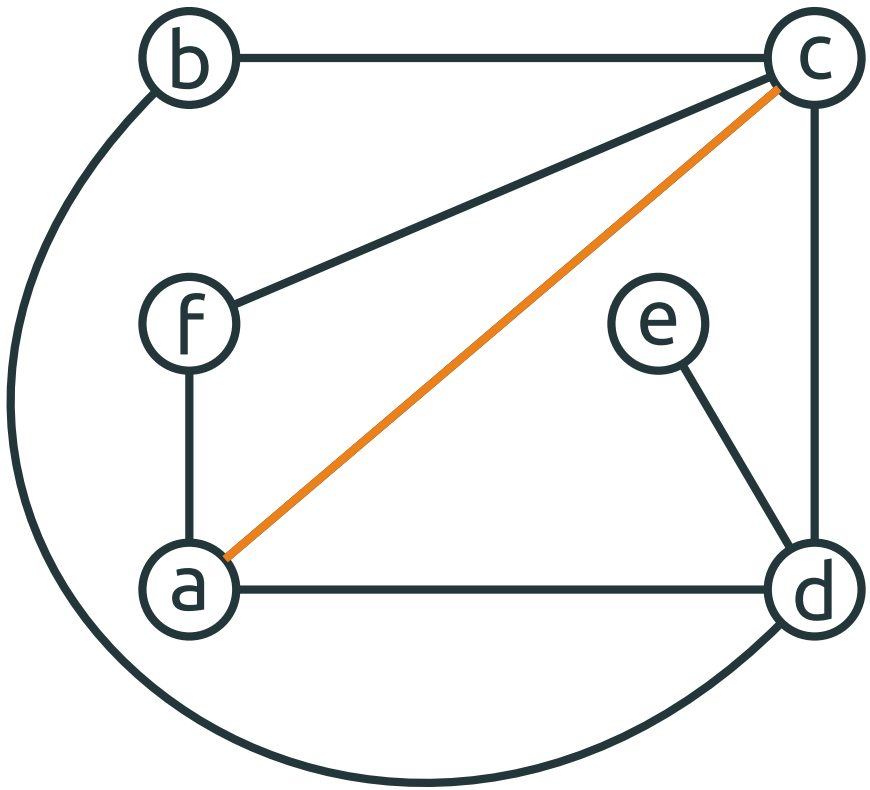
Faces and Edges

This face has 4 edges



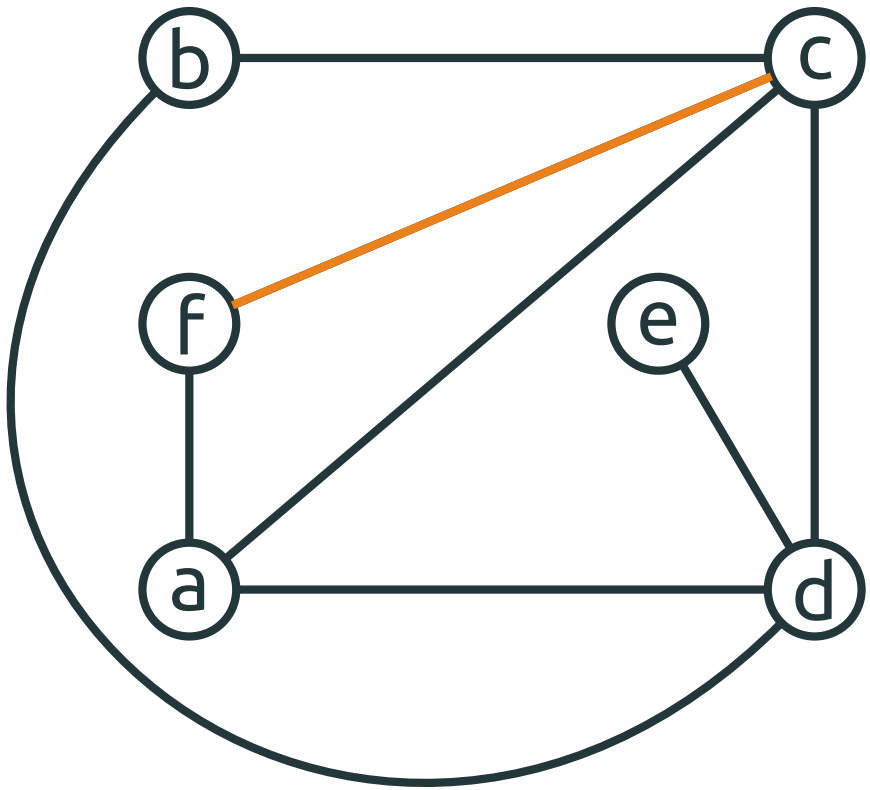
Faces and Edges

This edge belongs to 2 faces



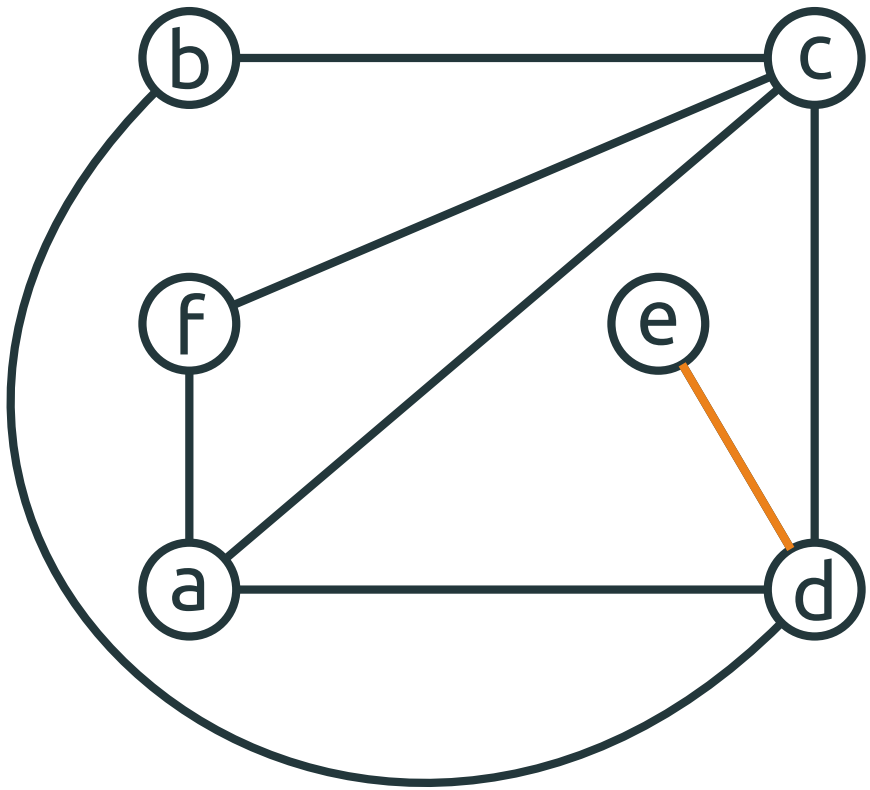
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Faces and Edges

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The Number of Faces

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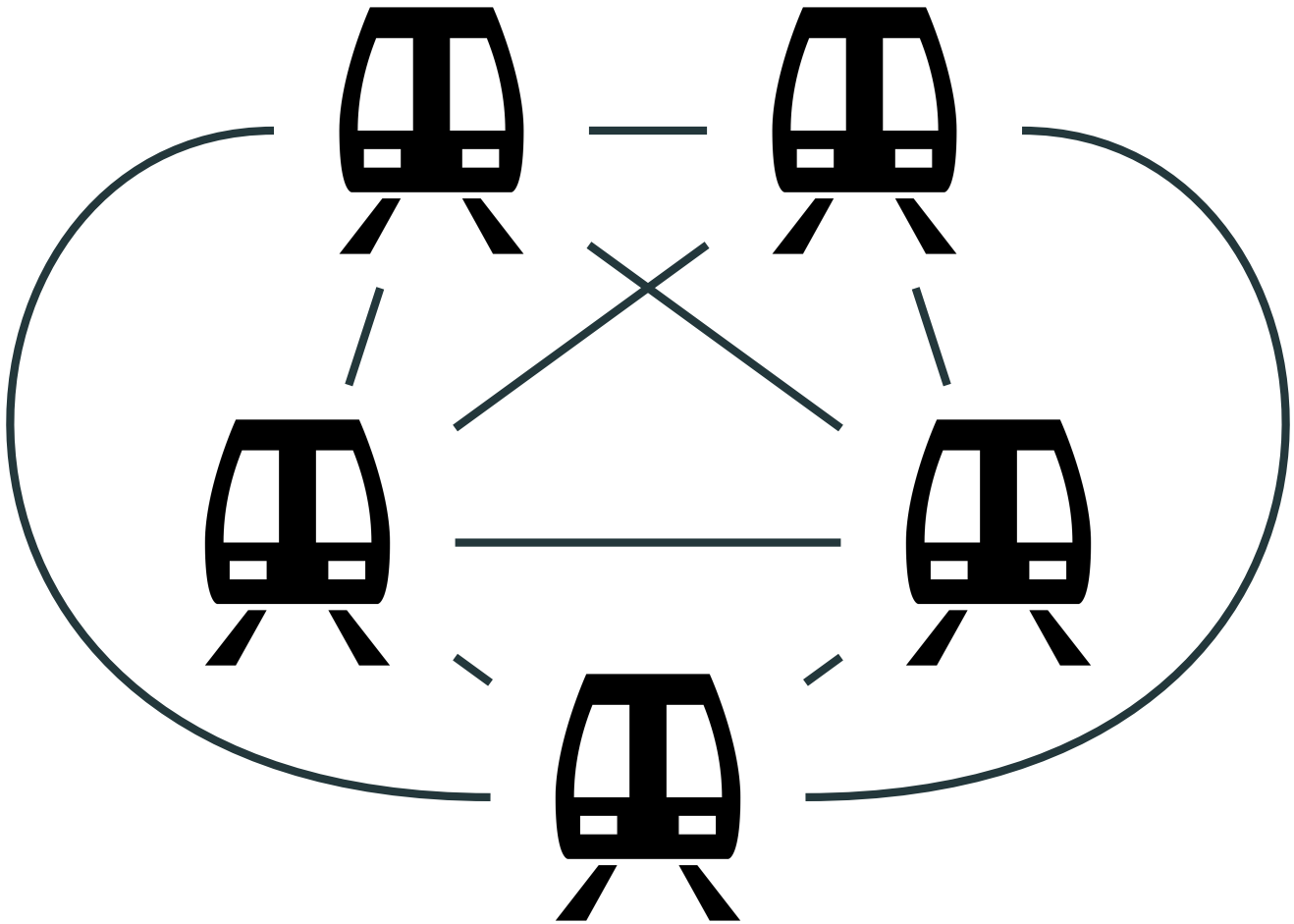
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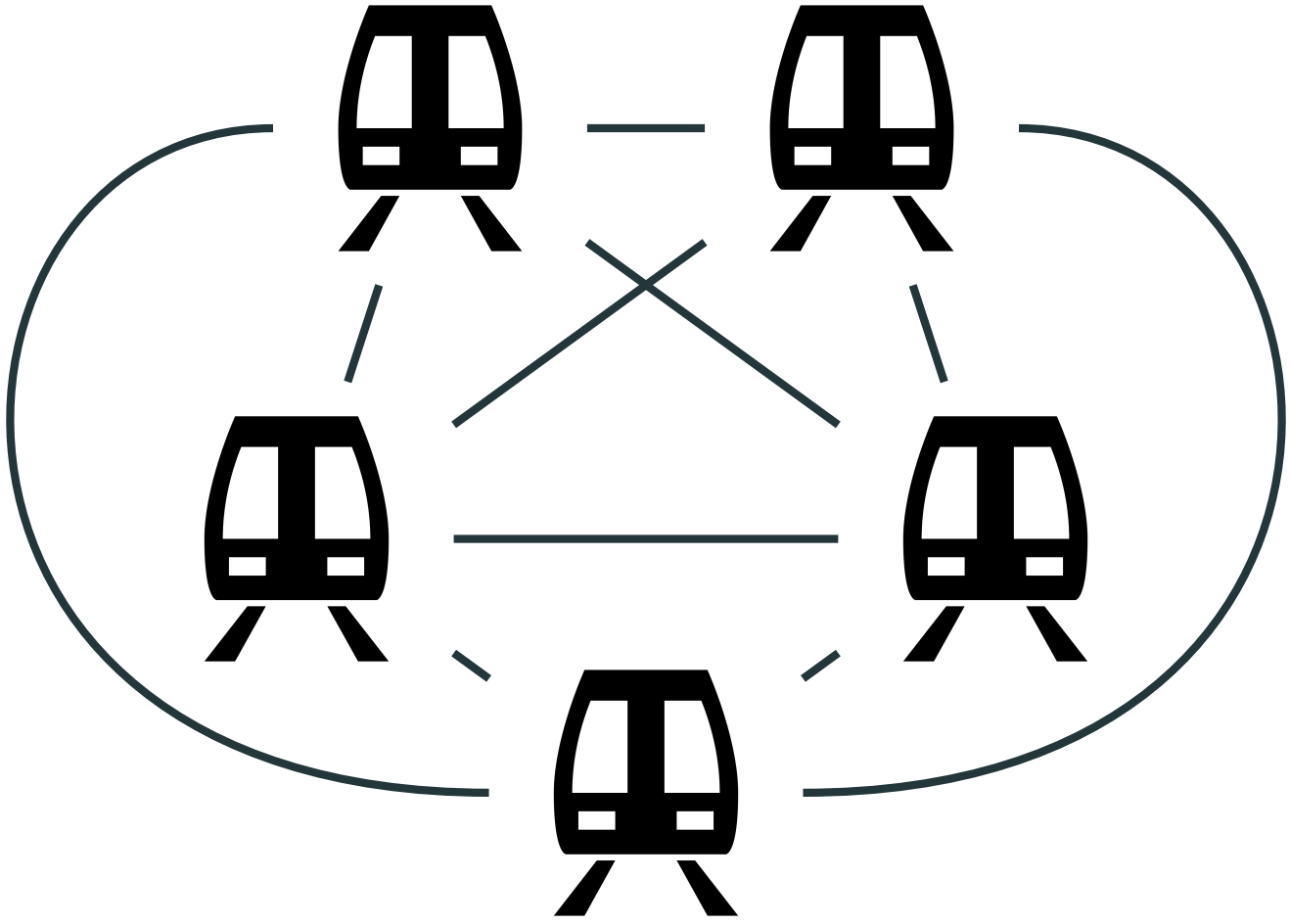
K_5 is Nonplanar

Why is K_5 nonplanar?



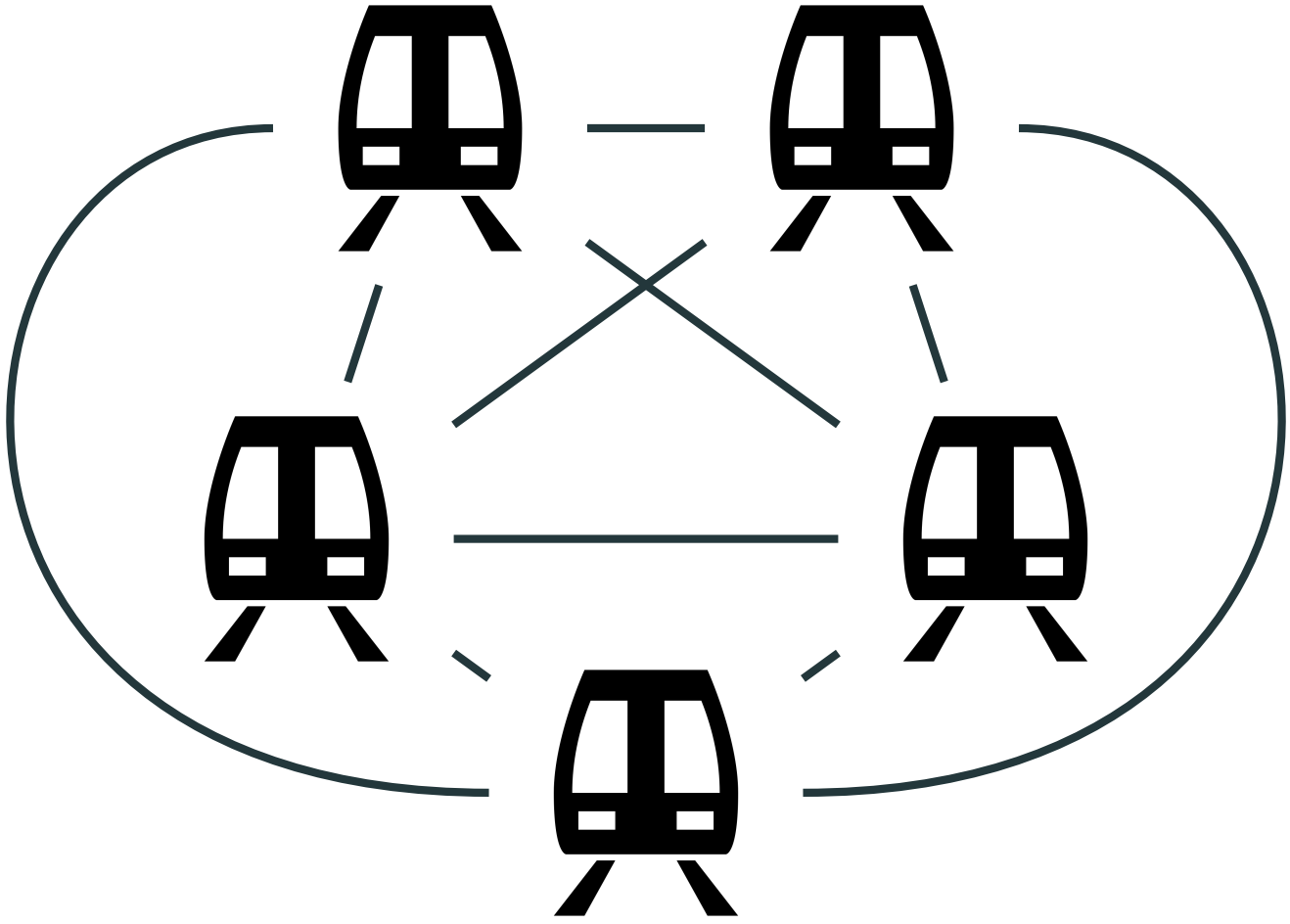
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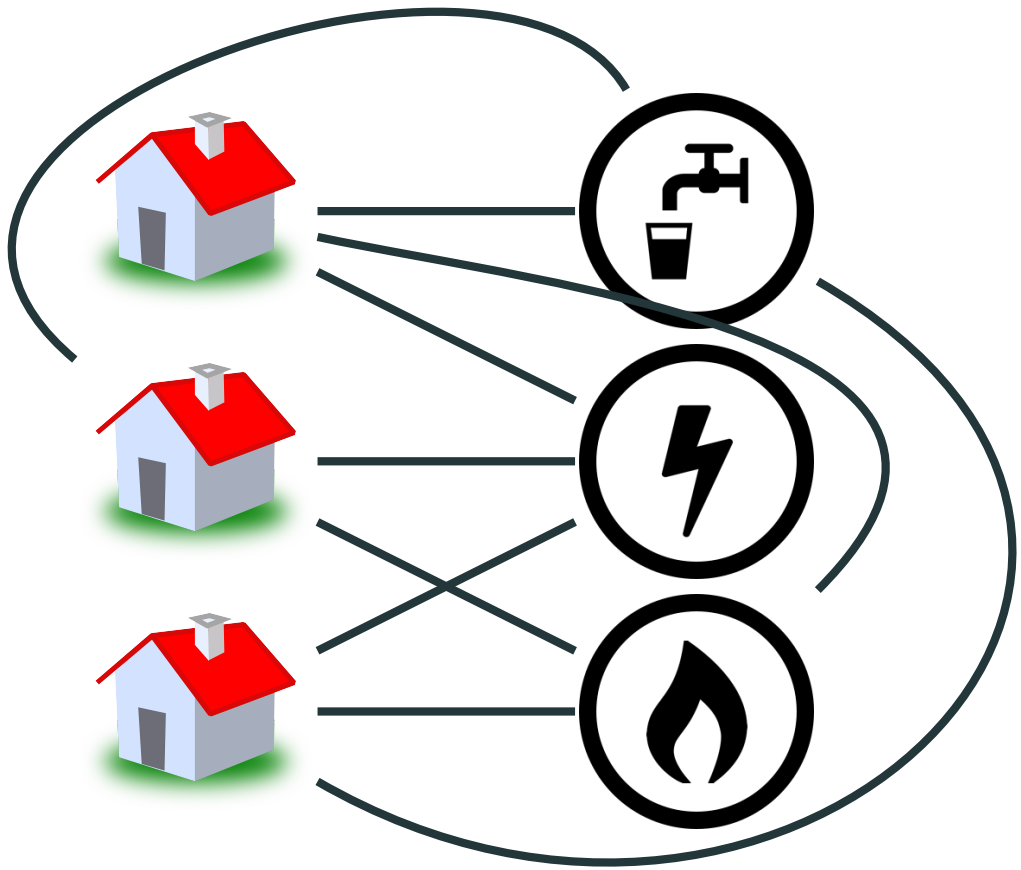


K_5 is Nonplanar

It has $v = 5$ vertices and $e = 10$ edges
In a planar graph, $e = 10$ must be $\leq 3v - 6 = 9$

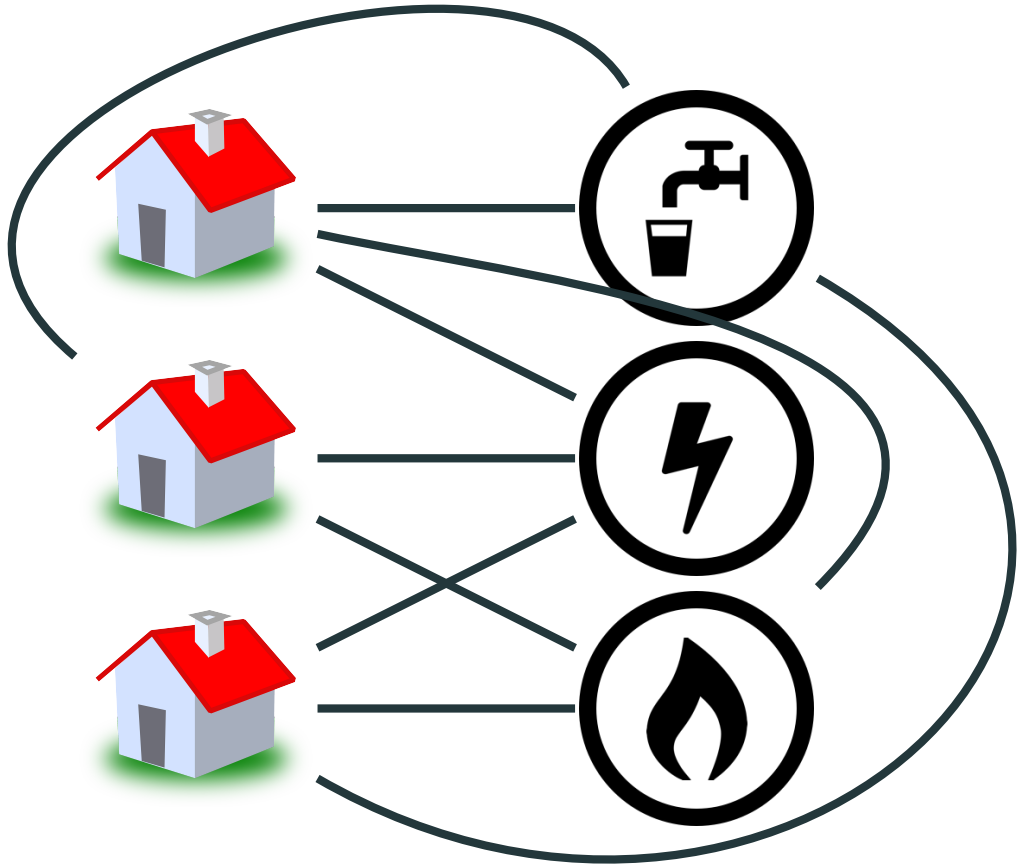


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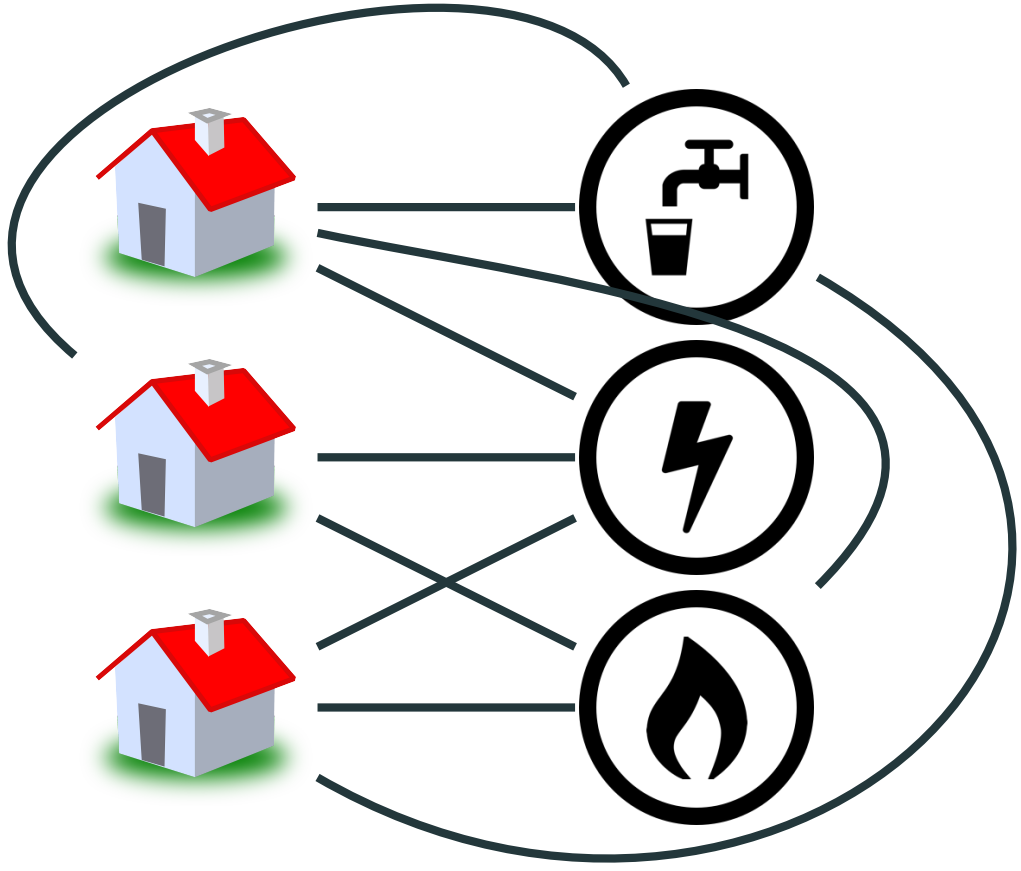
$$v = 6, e = 9$$



Is $K_{3,3}$ Planar?

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It does satisfy $e \leq 3v - 6$



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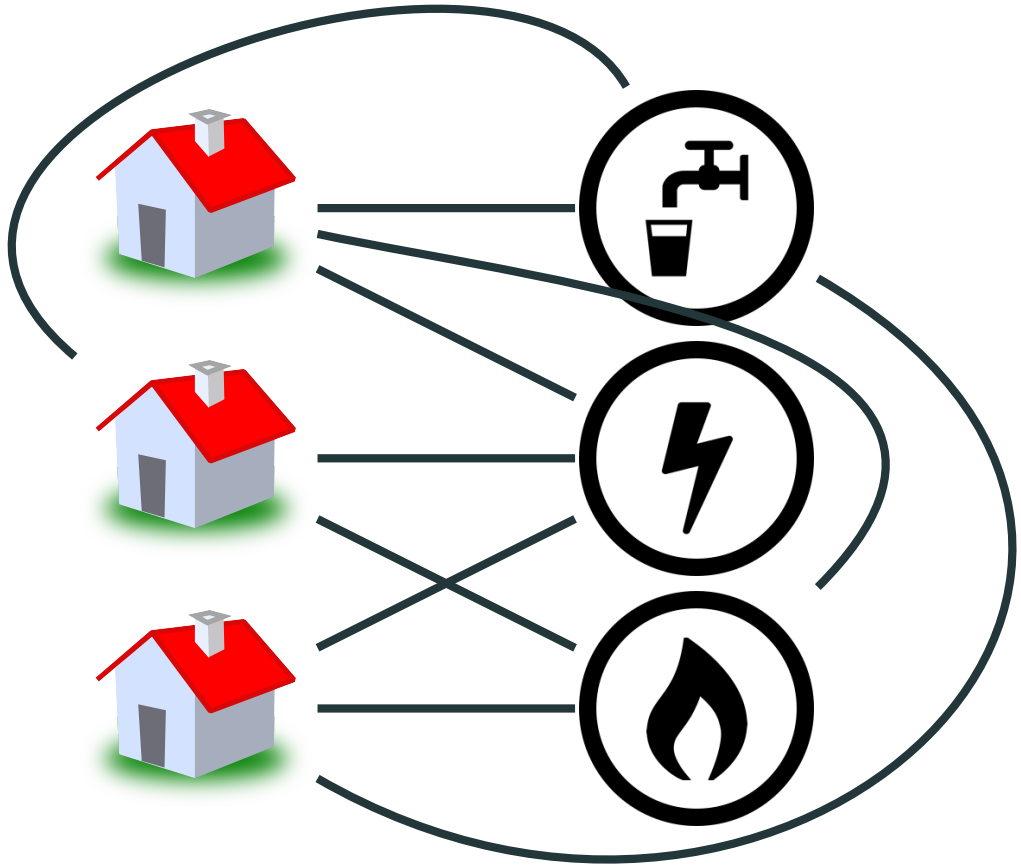
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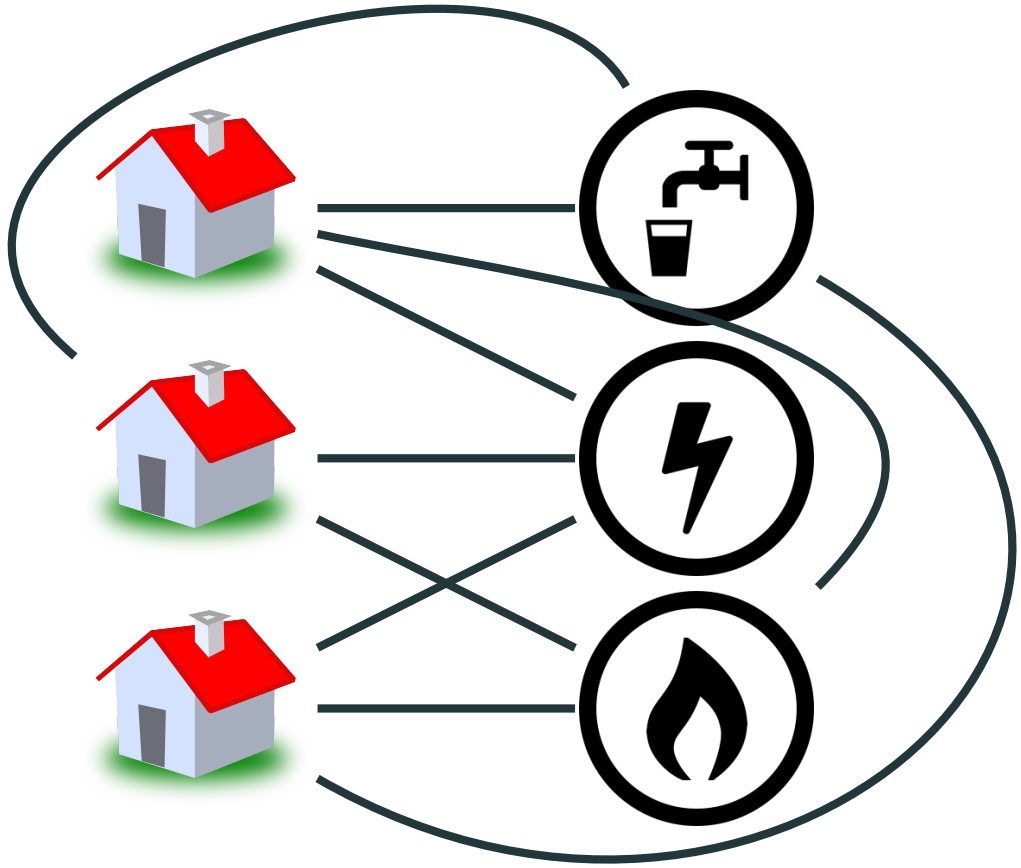
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In a planar bipartite graph,

$e = 9$ must be $\leq 2v - 4 = 8$

